Unified Geometric Landau EFT of Homogeneous $S^3 \times S^1$ Minisuperspace in Einstein–Cartan + Nieh–Yan Theory

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Abstract

We reconstruct the homogeneous $S^3 \times S^1$ minisuperspace reduction of Einstein–Cartan (EC) gravity augmented by the Nieh–Yan (NY) term, and demonstrate that the homogeneous geometric sector admits a unified description as a finite-degree-of-freedom geometric Landau effective theory. The aim is to organise the Pontryagin protection structure, the spin-2 shear deformations, and the spin-1 twist-mixing system along the irreducible spin-0/2/1 modes, thereby reconsolidating them into a single theoretical framework.

First, we show that the Pontryagin density $P = \langle R, \star R \rangle$ vanishes identically under the AX and VT torsion modes, while only the MX mode can sustain $P \neq 0$. In the MX mode the source at the isotropic point is

$$P_0 = \frac{2V\eta\,(V^2r^2 + 9\eta^2 - 36)}{9r^3},$$

establishing that non-zero Pontryagin number arises from torsion mixing $V\eta$, not from the twist field itself.

Second, we introduce the full shear tensor as the traceless symmetric tensor h_{ij} and organise its five components as the spin-2 quintet q_A. At the isotropic point Schur's lemma forces the spin-2 Hessian to

$$H_{AB} = \mu_q^2\,\delta_{AB}, \qquad \mu_q^2 = 48\pi^2,$$

a five-fold degeneracy. Under a squash background the $SO(3) \to SO(2)$ splitting reveals that the $m = 0$ singlet is the first soft mode on the positive-ε side at $\varepsilon_{c+} = 0.48324$, while on the negative side the $m = \pm 2$ sector undergoes a second-order transition into the $s \neq 0$ phase at $\varepsilon_{c-} = -0.29309 \pm 0.00019$.

Third, in the spin-1 sector the Hessian and field-space metric of the raw variables (ω_i, δ_i) are analysed as the generalised eigenvalue problem

$$Hv = \lambda Gv.$$

The field-space metric is rank-deficient; the null direction

$$Y_k \propto \left(1, \frac{2L}{r_0}\right)$$

is neither gauge nor variable redundancy, but a non-dynamical auxiliary direction. The physical spin-1 mode is

$$X_k = \frac{-2\omega_k + 3\delta_k}{\sqrt{13}},$$

with generalised eigenvalue $\lambda_{\text{phys}} = 4.1867 > 0$. A mixed-Hessian computation confirms that X_k fully decouples from the spin-2, radial, and torsion sectors at quadratic order.

Taking all results together, the model is understood as a unified geometric Landau-EFT on the homogeneous $S^3 \times S^1$ geometry, in which the homogeneous geometric degrees of freedom are organised along irreducible representations and the Pontryagin protection, phase-transition directions, physical/auxiliary decomposition, and allowed interaction terms are integrated into a single theoretical framework. This work provides one closed framework for the classical minisuperspace geometric sector in EC+NY theory and serves as a foundation for future extensions including quantisation.

1 Introduction

1.1 Background

In paper I [1] we compared $S^3 \times S^1$, $T^3 \times S^1$, and $\mathrm{Nil}^3 \times S^1$ within the Einstein–Cartan + Nieh–Yan (EC+NY) minisuperspace using the effective potential, and demonstrated the energetic dominance of the isotropic $S^3 \times S^1$ vacuum. In paper II [2] the Weyl extension and Pontryagin structure were added to that isotropic vacuum, and its stability was consolidated from both topological and dynamical perspectives.

The remaining open questions were the following.

1. In which sector is the Pontryagin density genuinely generated?
2. Can squash/shear deformations be treated not as auxiliary coordinates but as a complete spin-2 multiplet?
3. Can the six components of twist and mixing be uniquely decomposed into a physical mode and a non-dynamical auxiliary mode?
4. Can all of the above be unified into a single effective theory?

This paper aims to consolidate these four points as a single *completed homogeneous geometric sector*.

1.2 Viewpoint of This Paper

We first state explicitly the basis adopted throughout. The geometric degrees of freedom are ultimately reorganised as follows:

- **spin-0**: radial mode r and torsion scalars (η, V),
- **spin-2**: traceless symmetric quintet q_A,
- **spin-1**: physical triplet X_k and auxiliary triplet Y_k.

The single squash variable ε introduced in paper II [2] and the shear variable s introduced in the present work are ultimately understood as parts of the spin-2 quintet.

1.3 Main Results

The main results of this paper are summarised in the following five points.

1. **Pontryagin protection theorem**: $P \equiv 0$ holds exactly in the AX/VT modes; only the MX mode supports $P \neq 0$.

2. **Spin-2 completion**: The full shear tensor is completed as the spin-2 quintet, with five-fold degeneracy $\mu_q^2 = 48\pi^2$ at the isotropic point.

3. **Final phase diagram**: On the positive side the $m = 0$ singlet softens at $\varepsilon_{c+} = 0.48324$; on the negative side the $m = \pm 2$ sector undergoes a second-order transition into the $s \neq 0$ phase at $\varepsilon_{c-} = -0.29309 \pm 0.00019$.

4. **Spin-1 physical/auxiliary decomposition**: Y_k is neither gauge nor redundancy but a non-dynamical auxiliary; $X_k = (-2\omega_k + 3\delta_k)/\sqrt{13}$ provides the unique propagating spin-1 basis.

5. **Unified Landau-EFT**: In the final basis the quadratic sector is organised block-diagonally, and the allowed cubic couplings and forbidden terms are systematically classified by representation theory.

1.4 Outline

The paper is organised as follows.

- Section 2 defines the action, homogeneous ansatz, and adopted variables.
- Section 3 establishes the Pontryagin protection theorem and the MX exception.
- Section 4 discusses the full shear quintet, $SO(2)$ splitting, and phase-transition structure.
- Section 5 reconstructs the spin-1 sector as a generalised eigenvalue problem and establishes the physical/auxiliary decomposition.
- Section 6 summarises the final Landau-EFT and the mode dictionary.
- Section 7 clarifies what has been completed and what remains for future work.
- Section 8 is the conclusion.
- Appendices A–H contain the computational pipeline, proofs, mass-splitting formulae, cross-Hessian details, invariant basis, and numerical tables.

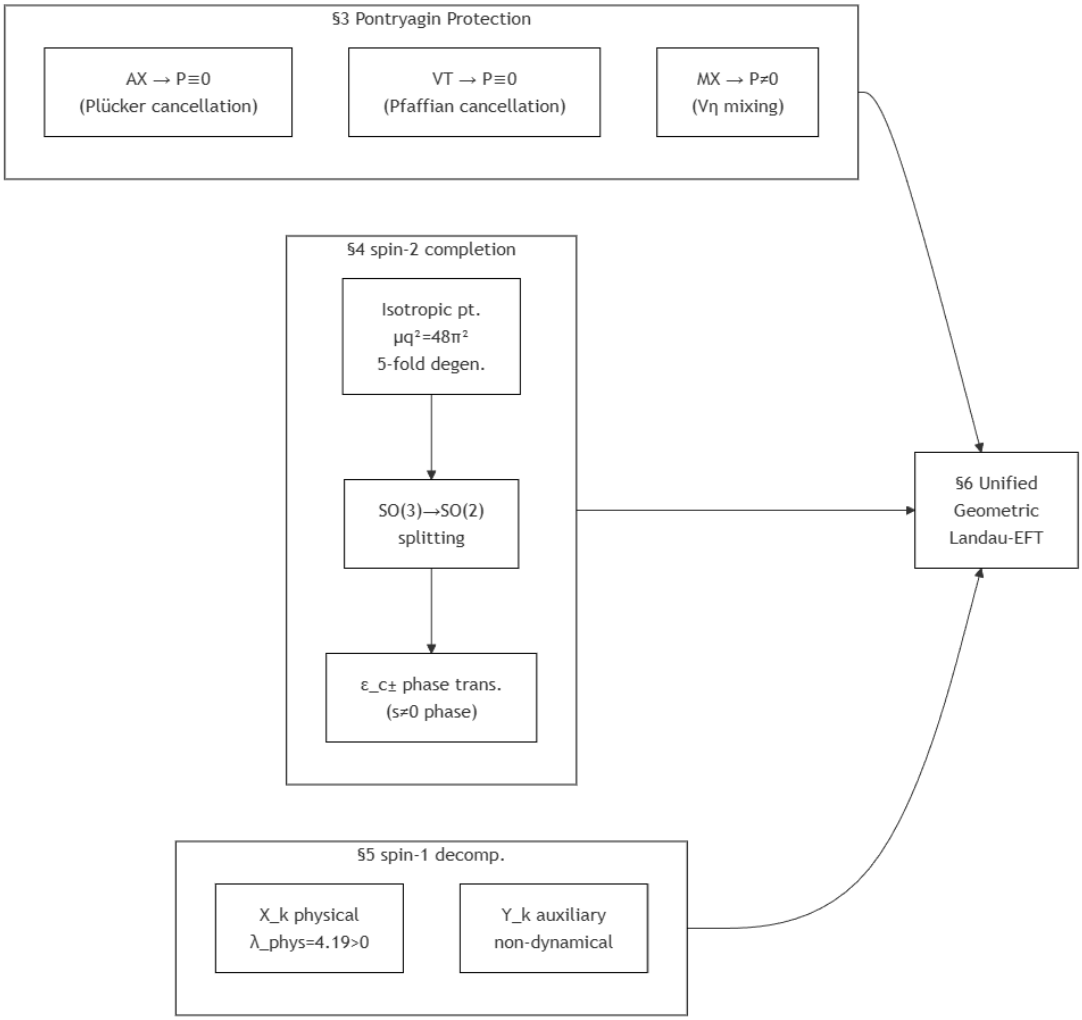


Figure 1: Logical structure of this paper and connections between sections. The three pillars—Pontryagin protection (§3), spin-2 completion (§4), and spin-1 decomposition (§5)—converge into the unified geometric Landau-EFT of §6.

2 Setup and Geometric Framework

This section collects the degrees of freedom, action, and geometric conventions. We adopt the final basis as the canonical description throughout.

2.1 Effective Potential

The effective potential in the homogeneous minisuperspace reduction is

$$V_{\text{eff}} = -\left(\frac{R}{2\kappa^2} + \theta_{\text{NY}} N\right) \text{Vol}, \qquad \text{Vol} = 2\pi^2 L r^3, \tag{1}$$

where R is the Ricci scalar computed from the Einstein–Cartan connection [3], N is the Nieh–Yan density [4], r is the S^3 radius, and L is the S^1 scale. In subsequent isotropic-point evaluations we normalise $L = 1$, at which the equilibrium radius is $r_0 = 3$.

The Pontryagin density is defined by

$$P = \langle R, \star R \rangle = \frac{1}{2}\, \varepsilon^{abcd} R_{abef} R_{cd}{}^{ef}. \tag{2}$$

The central subject of this paper is the organisation of V_{eff} and P in the final basis of homogeneous geometric freedoms.

2.2 Homogeneous Ansatz and Spatial Frame

The spatial 3-frame is written using left-invariant 1-forms σ^i as

$$e^i = r\, M^i{}_j\, \sigma^j, \tag{3}$$

where $M^i{}_j = (\exp h)^i{}_j$ with $h = \sum_A q_A T_A$ the full shear tensor (see §2.3). Full shear deformations are embedded as constant frame deformations. The complete coframe—including the twist ω_k and spin-1 mixing δ_k definitions—is given in Appendix B.

Under a constant frame deformation the structure constants transform as

$$C'^i{}_{jk} = M^i{}_a\, (M^{-1})^b{}_j\, (M^{-1})^c{}_k\, C^a{}_{bc}, \tag{4}$$

and the homogeneous Maurer–Cartan system is preserved.

2.3 Spin-2 Quintet: Full Shear Basis

The traceless symmetric tensor h_{ij} is expanded as

$$h_{ij} = \sum_{A=1}^{5} q_A\, (T_A)_{ij}, \qquad \text{tr}(T_A T_B) = \delta_{AB}. \tag{5}$$

The adopted basis matrices are

$$T_1 = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}, \qquad T_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \tag{6}$$

$$T_3 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad T_4 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad T_5 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}. \tag{7}$$

The fundamental invariants in this basis are

$$I_2 = \mathrm{tr}(h^2) = \sum_{A=1}^{5} q_A^2, \qquad I_3 = h_{ij}\omega_i\omega_j, \qquad I_4 = \mathrm{tr}(h^3), \qquad W_2 = \omega_i\omega_i. \tag{8}$$

An exact check confirms that the component expansion $I_3 = q_A F_A(\omega)$ holds under the adopted T_A conventions.

2.4 Spin-1 Variables and Torsion Modes

The raw geometric variables of the spin-1 sector are the twist ω_i and the mixing δ_i. The raw basis is not, however, the final physical basis. The generalised eigenvalue analysis of §5 decomposes the raw variables into the physical propagating triplet X_k and the non-dynamical auxiliary Y_k. Explicit expressions and derivations are given in §5.

For the torsion scalars we distinguish three modes:

- AX mode: $(\eta \neq 0,\ V = 0)$,
- VT mode: $(\eta = 0,\ V \neq 0)$,
- MX mode: $(\eta \neq 0,\ V \neq 0)$.

This distinction is essential for the Pontryagin structure.

2.5 Representation-Theoretic Organisation

The degrees of freedom are organised by $SO(3)$ irreducible representations as follows.

Sector	Variable	DOF	Representation
Radial	r	1	spin-0
Shear	q_A	5	spin-2
Raw spin-1	(ω_i, δ_i)	6	spin-1 $\oplus$ spin-1
Physical spin-1	X_k	3	spin-1
Auxiliary spin-1	Y_k	3	spin-1
Torsion	(η, V)	2	spin-0

Table 1: $SO(3)$ irreducible decomposition of the homogeneous geometric degrees of freedom. The raw spin-1 variables (ω_i, δ_i) are an intermediate description; the final physical dictionary uses X_k and Y_k, established in §5.

In particular, ε and s are ultimately understood as particular projections of q_A and are not treated as standalone irreducible variables. The physical/auxiliary classification of the spin-1 sector is established by the generalised eigenvalue analysis of §5, and the definitive mode dictionary is presented in §6.

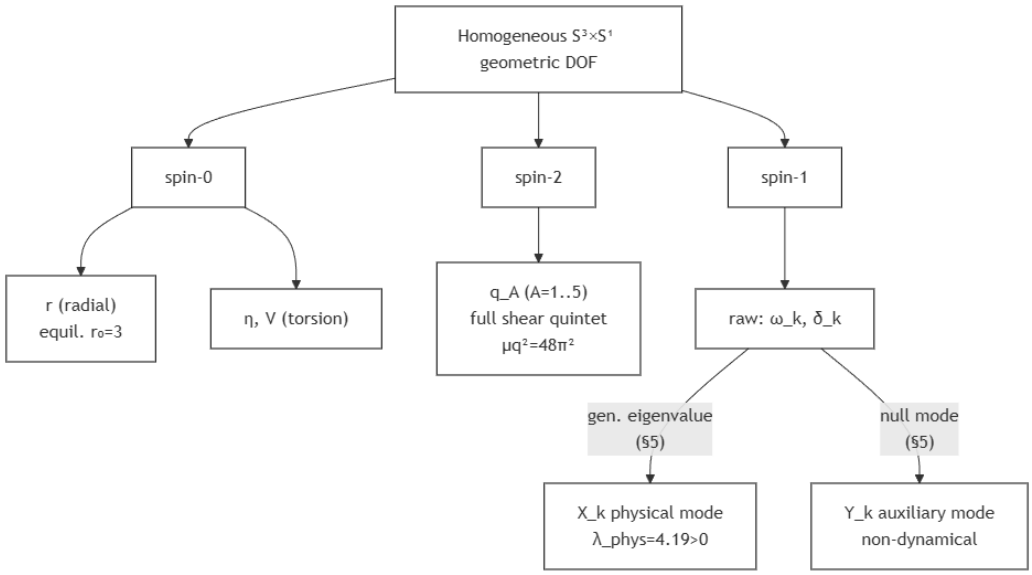


Figure 2: $SO(3)$ irreducible decomposition of the homogeneous $S^3 \times S^1$ geometric degrees of freedom and the final mode classification. The raw spin-1 variables are decomposed into the physical triplet X_k ($\lambda_{\text{phys}} = 4.19 > 0$) and the non-dynamical auxiliary Y_k via the generalised eigenvalue analysis of §5.

2.6 Scope

This paper addresses the homogeneous minisuperspace geometric sector. Consequently, local inhomogeneous modes, boundary conditions, and full field-theoretic gauge fixing are outside the scope of the present work and are deferred to future investigations (see §7).

3 Pontryagin Protection Structure

The message of this section is simple. The Pontryagin density P is not generated everywhere in the homogeneous sector. It vanishes exactly in the AX and VT modes, and only the MX mode is an exception. This structure is established not by a qualitative symmetry argument but by algebraic cancellations in the Riemann blocks.

3.1 Pontryagin Density and Block Decomposition

The Pontryagin density $P = \langle R, \star R \rangle$ is as defined in §2. In the 3+1 block notation it reads

$$P = \sum_{c<d} \Big(R_{01cd} R_{23cd} - R_{02cd} R_{13cd} + R_{12cd} R_{03cd} \Big), \tag{9}$$

enabling us to track the cancellation structure for each pair (c, d).

3.2 Theorem P0: AX and VT Sectors

Theorem 1 (P0)**.** *For the AX and VT modes and for the general vector-twist ansatz on* $S^3 \times S^1$, $\mathrm{Nil}^3 \times S^1$, *and* $T^3 \times S^1$, $P \equiv 0$ *holds identically.*

For isotropic twist on S^3 and Nil^3, the individual Plücker contribution at each fixed (c, d),

$$R_{01cd} R_{23cd} - R_{02cd} R_{13cd} + R_{12cd} R_{03cd} = 0, \tag{10}$$

vanishes independently, so

$$P = \sum_{c<d} \Big(R_{01cd} R_{23cd} - R_{02cd} R_{13cd} + R_{12cd} R_{03cd} \Big) = 0. \tag{11}$$

The key point is that this is *not* an accidental cancellation in the sum; each (c, d) term vanishes independently. Furthermore, symbolic computation confirms $P \equiv 0$ for anisotropic vector twist as well, so the protection mechanism does not depend on isotropy per se.

3.3 VT Sector: Pfaffian Cancellation

In the VT mode the cancellation mechanism is more refined.

Lemma 1 (VT-Pfaffian)**.** *In the VT mode, each Riemann-block Pfaffian is individually non-zero, yet the SS group and the S3 group cancel with equal magnitude and opposite sign, yielding* $P \equiv 0$ *overall.*

This cancellation holds for both $S^3 \times S^1$ and $\mathrm{Nil}^3 \times S^1$. The VT-mode protection is a *Pfaffian group cancellation between the SS and S3 classes*, not a trivial decomposability of

individual blocks. Explicit Pfaffian expressions and verification for both topologies are given in Appendix B.3.

3.4 MX Sector as the Unique Source of $P \neq 0$

Proposition 1 (MX-P). *In the MX mode the Pontryagin density is generated in proportion to the torsion mixing $V\eta$; the twist field itself is not the origin.*

For $S^3 \times S^1$,

$$P(\boldsymbol{\omega}) = \underbrace{\frac{2V\eta\,(V^2r^2 + 9\eta^2 - 36)}{9r^3}}_{P_0 = P(\omega=0)} + \underbrace{\frac{40V\eta L^2}{3r^5}}_{C_1} I_2. \tag{12}$$

Even at the isotropic point one has

$$P_0 = \frac{2V\eta\,(V^2r^2 + 9\eta^2 - 36)}{9r^3}, \tag{13}$$

so the origin of non-zero Pontryagin number is vacuum torsion mixing. One should therefore not identify chirality generation in the MX mode with spontaneous twist generation.

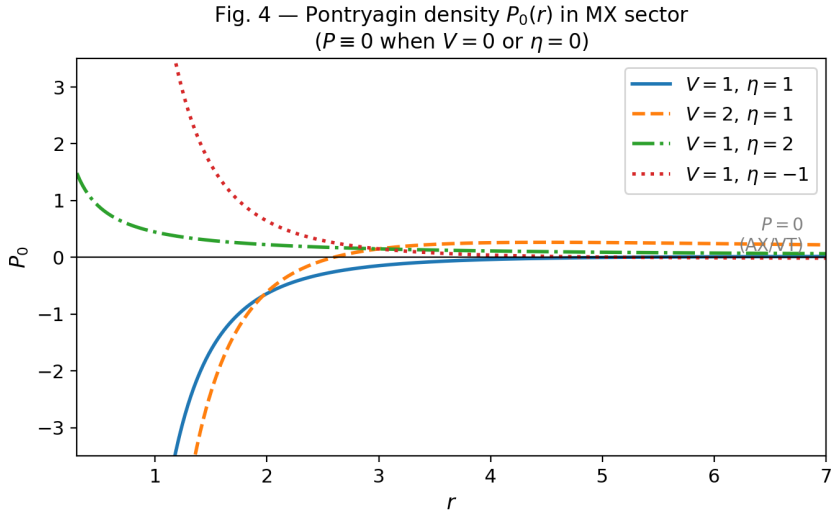


Figure 3: Pontryagin density P in the MX sector as a function of the torsion mixing parameter $V\eta$. Non-zero P arises exclusively from vacuum torsion mixing; the twist field ω is not the origin.

3.5 NyVariant Independence and Twist Stability

Because the Pontryagin density depends on the Riemann tensor, it is independent of the NyVariant choice (FULL/TT/REE): $P \equiv 0$ always in AX/VT, and always non-zero in MX. A summary table is given in Appendix B.5.

Within the parameter range analysed in this paper, spontaneous twist generation does

not occur either. From the isotropic twist effective potential on $S^3 \times S^1$,

$$m^2_{S^3}(r) = \left.\frac{\partial^2 V_{\text{eff}}}{\partial \omega^2}\right|_{\omega=0} = \frac{48\pi^2 L^3}{\kappa^2 r} > 0, \tag{14}$$

so $\omega = 0$ is stable for all $r > 0$ [2].

3.6 Interpretation

The conclusion of this section is stated precisely in three points.

1. The AX and VT modes are structurally protected.
2. The MX mode is the unique exception, with origin $V\eta$ mixing [5].
3. Within the scope of this paper, the twist sector is itself stable and no spontaneous twist generation occurs.

This protection structure is the unified starting point that is consistent with the spin-2 and spin-1 organisation introduced below.

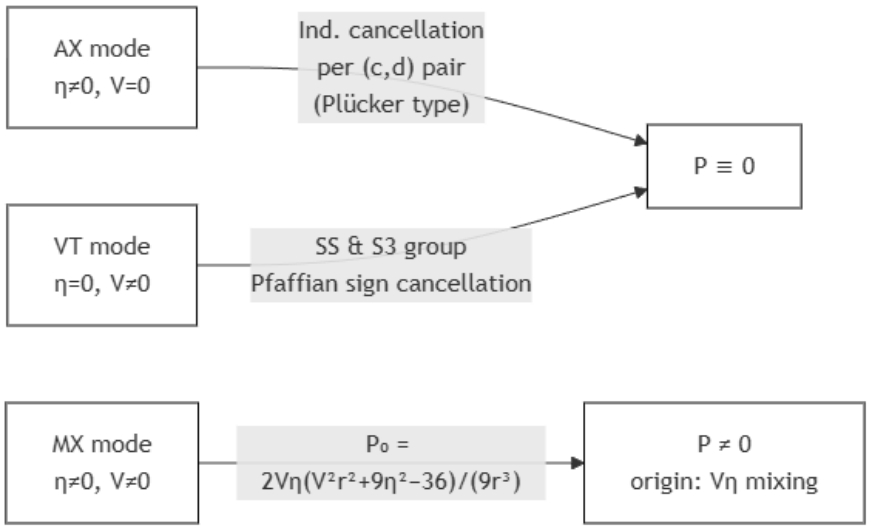


Figure 4: Classification of the Pontryagin density by torsion mode. The AX and VT modes are protected by distinct algebraic mechanisms; the MX mode is the unique exception sourced by vacuum torsion mixing $V\eta$.

4 Spin-2 Sector: Full Shear Completion and Geometric Phase Structure

This section completes the squash/shear deformations as the full shear tensor and presents the final phase diagram of the spin-2 quintet. The conclusions are: five-fold degeneracy at the isotropic point; softening of the $m = 0$ singlet on the positive-ε side; and a second-order phase transition of the $m = \pm 2$ sector on the negative-ε side.

4.1 Full Shear Tensor as the Spin-2 Quintet

A description in terms of raw geometric variables (ε, s, ϕ) is useful for following a particular slice but insufficient to represent the entire spin-2 sector. We adopt

$$h_{ij} = \sum_{A=1}^{5} q_A \, (T_A)_{ij} \tag{15}$$

and treat the five components of $\mathrm{Sym}_0^2(\mathbb{R}^3)$ as a complete spin-2 quintet. Here q_A are the final variables aligned with the $SO(3)$ irreducible basis; ε and s are used as projections of q_A onto specific directions.

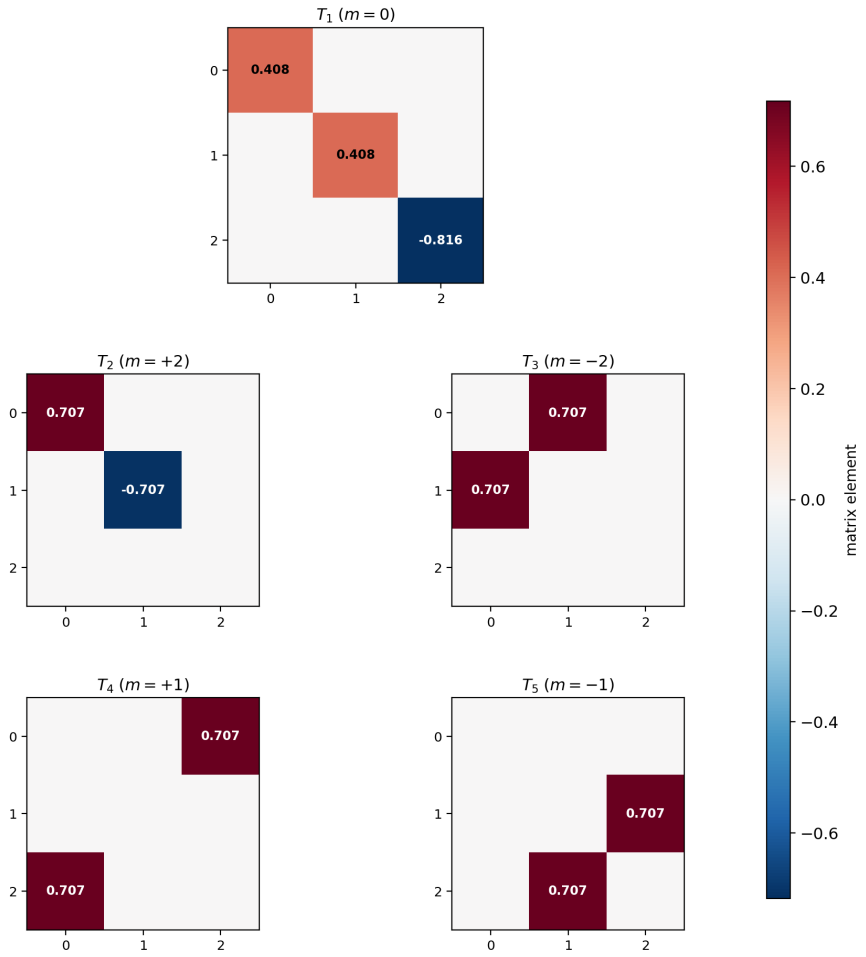


Figure 5: Basis vectors $T_1, \ldots, T_5$ of the spin-2 quintet $\mathrm{Sym}^2_0(\mathbb{R}^3)$. The three sectors correspond to the residual $SO(2)$ representations $m = 0$, $m = \pm 2$, and $m = \pm 1$.

4.2 Isotropic Point and Schur Degeneracy

At the isotropic point the residual symmetry is the full $SO(3)$, so the spin-2 Hessian is proportional to the identity matrix on each irreducible representation. Consequently,

$$V_{\text{eff}} = V_0 + \lambda\,\mathrm{tr}(h^2) + O(|q|^3) = V_0 + \lambda \sum_{A=1}^{5} q_A^2 + O(|q|^3), \tag{16}$$

giving

$$H_{AB} = \left.\frac{\partial^2 V}{\partial q_A \partial q_B}\right|_0 = 2\lambda\,\delta_{AB}. \tag{17}$$

The generalised mass is equal for all five components:

$$\mu_q^2 = 48\pi^2. \tag{18}$$

This is the spin-2 sector's Schur degeneracy.

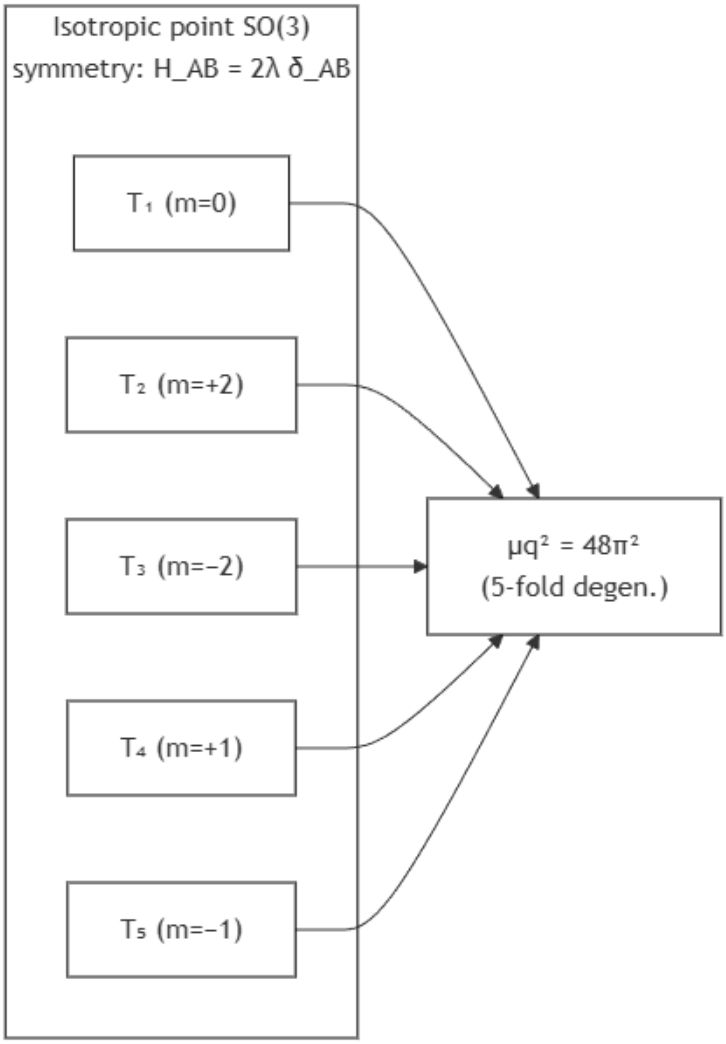


Figure 6: Five-fold degeneracy of the spin-2 quintet at the isotropic point. Schur's lemma $H_{AB} = 2\lambda\delta_{AB}$ enforces equal mass $\mu_q^2 = 48\pi^2$ for all five components.

4.3 $SO(3)$ to $SO(2)$ Splitting under Squash Background

Under a positive-ε squash background the symmetry reduces to $SO(3) \to SO(2)$, and the spin-2 quintet splits as follows.

Residual charge	Basis	Interpretation
$m = 0$	T_1	squash singlet
$m = \pm 2$	(T_2, T_3)	in-plane doublet
$m = \pm 1$	(T_4, T_5)	off-axis doublet

Table 2: Residual $SO(2)$ decomposition of the spin-2 quintet under a positive-ε squash background.

The key observation is that the order in which soft modes appear is organised by these residual representations.

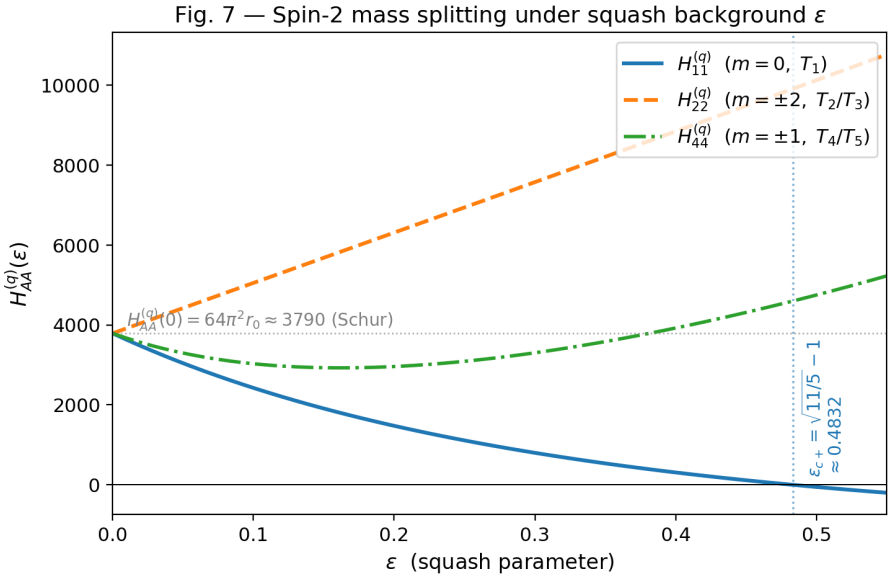


Figure 7: Mass splitting of the spin-2 quintet as a function of the squash parameter ε. The three branches correspond to $m = 0$ (singlet T_1), $m = \pm 2$ (doublet T_2, T_3), and $m = \pm 1$ (doublet T_4, T_5). The $m = 0$ branch reaches zero first at ε_{c+}.

4.4 Positive-ε Side: First Soft Mode

From the full shear analysis, on the positive side the T_1 singlet becomes soft first. The critical value is

$$\varepsilon_{c+} = \sqrt{\frac{11}{5}} - 1 \approx 0.48324, \qquad r_0(\varepsilon_{c+}) \approx 2.8114, \tag{19}$$

at which $\mu_\varepsilon^2 \to 0$.

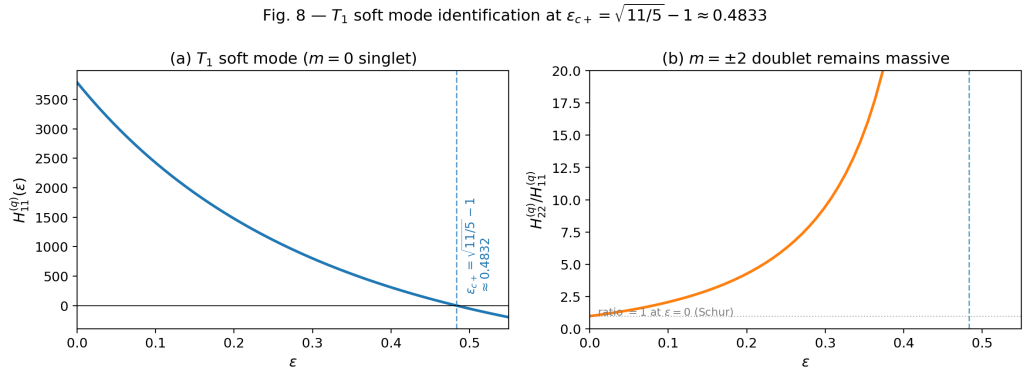


Figure 8: Softening of the $m = 0$ mode $\mu_\varepsilon^2(\varepsilon)$ on the positive-ε side. The curve reaches zero at the exact critical point $\varepsilon_{c+} = \sqrt{11/5} - 1 \approx 0.483$, while the $m = \pm 2$ branch remains massive.

Key points to emphasise:

- The first soft mode is the $m = 0$ singlet T_1.

- The $m = \pm 2$ doublet (T_2, T_3) remains massive even at ε_{c+}.
- A global scan over (r, ε) revealed no new global minimum; no condensation was found on the axisymmetric slice.

Therefore ε_{c+} is a *softening threshold* on the positive side rather than the birth point of a fully condensed phase.

4.5 Negative-ε Side: $s \neq 0$ Phase Transition

On the negative side the situation differs. A three-indicator analysis,

$$a(\varepsilon) = \left.\frac{d^2F}{ds^2}\right|_{s=0}, \qquad s_0(\varepsilon) = \arg\min_s F(s;\varepsilon), \qquad \mu_s^2(\varepsilon) = \frac{H_{ss}}{G_{ss}}, \tag{20}$$

locates the critical point at

$$\varepsilon_{c-} = \frac{1}{\sqrt{2}} - 1 \approx -0.29309 \pm 0.00019, \tag{21}$$

with the three indicators converging to -0.29303, -0.29335, and -0.29289, respectively. The closed-form exact value $1/\sqrt{2} - 1$ agrees with the numerical estimate and is derived directly from the condition $H_{22}^q = 0$ (see Appendix D).

For the Landau free energy

$$F(s;\varepsilon) = a(\varepsilon)\, s^2 + b(\varepsilon)\, s^4 + O(s^6), \tag{22}$$

one finds $b(\varepsilon) \in [2288, 7805] > 0$ near the critical point, confirming a *second-order transition*. The order-parameter fit gives

$$s_0 \propto |\varepsilon - \varepsilon_c|^\beta, \qquad \beta = 0.4678, \quad R^2 = 0.987, \tag{23}$$

consistent with a mean-field-like continuous transition. β falls slightly below the mean-field value $1/2$ by $1/2 - \beta \approx 0.032$; the precise physical meaning of this deviation—which is sensitive to the finite-range fit, higher-order terms, and the finite precision of the minisuperspace Landau description—is beyond the scope of this paper.

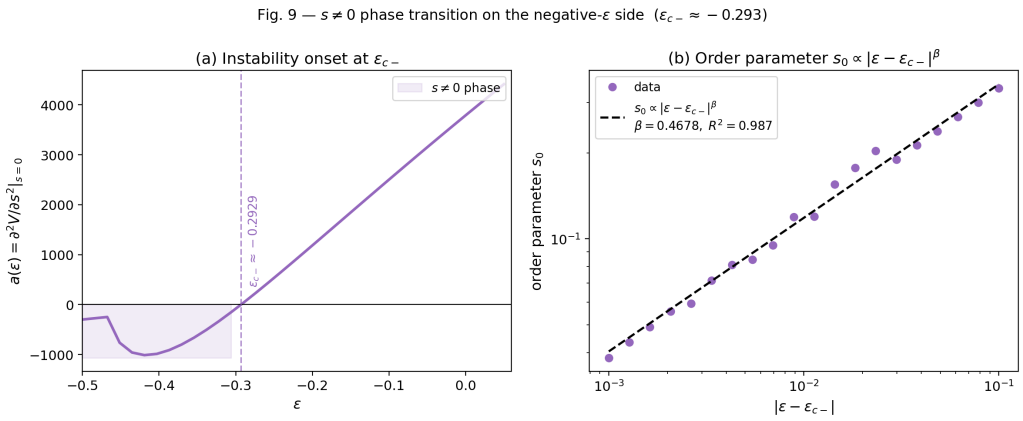


Figure 9: Phase structure on the negative-ε side. The order parameter $s_0(\varepsilon)$ emerges continuously at $\varepsilon_{c-} = 1/\sqrt{2} - 1 \approx -0.293$, consistent with a second-order transition into the $s \neq 0$ phase.

4.6 Geometric Interpretation

In the $s \neq 0$ vacuum, μ_s^2 becomes soft near the transition and then returns to positive values, while μ_ε^2 remains large and positive over a wide range. Furthermore, the ϕ direction is a pseudo-Goldstone rather than an exact Goldstone, reflecting that $SO(2)$ is explicitly broken in the $s \neq 0$ background.

The final picture of the spin-2 sector is summarised as follows.

1. The full shear 5-plet is five-fold degenerate at $\mu_q^2 = 48\pi^2$ at the isotropic point.
2. The first unstable direction on the positive side is the $m = 0$ singlet T_1.
3. On the negative side the $m = \pm 2$ sector governs the second-order transition into the $s \neq 0$ phase.

This phase diagram plays the central role of the spin-2 block in the Landau-EFT description of §6.

5 Spin-1 Sector: Twist-Mixing System and Physical/Auxiliary Decomposition

This section replaces the raw geometric variables (ω_i, δ_i) with the physical basis and reconstructs the spin-1 sector via the generalised eigenvalue problem

$$Hv = \lambda G v. \tag{24}$$

The conclusion is that Y_k is a non-dynamical auxiliary direction and X_k is the unique propagating spin-1 triplet.

5.1 Raw Spin-1 Variables

At the isotropic point $r_0 = 3$, $L = 1$, the raw 2×2 blocks are

$$H_2 = \begin{pmatrix} \mu_\omega^2 & m_{\omega\delta} \\ m_{\omega\delta} & \mu_\delta^2 \end{pmatrix} = \begin{pmatrix} 52.6379 & -631.6544 \\ -631.6544 & 3789.9295 \end{pmatrix}, \tag{25}$$

$$G_2 = \begin{pmatrix} G_{\omega\omega} & G_{\omega\delta} \\ G_{\omega\delta} & G_{\delta\delta} \end{pmatrix} = \begin{pmatrix} 236.870435 & -355.305547 \\ -355.305547 & 532.958162 \end{pmatrix}. \tag{26}$$

Looking only at the eigenvalues of the raw Hessian gives the impression of a negative direction, but this is insufficient for determining physical mass. The physically meaningful quantity is the generalised eigenvalue problem for the pair (H, G).

5.2 Null Direction and Physical Direction

The field-space metric G is rank-deficient. For each axis k it has a null direction

$$Y_k \propto (\omega_k, \delta_k) = \left(1, \frac{2L}{r_0}\right) = \left(1, \frac{2}{3}\right), \qquad k = 0, 1, 2, \tag{27}$$

where $k = 0, 1, 2$ labels the three spin-1 axes and is distinct from the spatial tensor indices $i, j = 1, 2, 3$. The physical direction orthogonal to Y_k is

$$X_k = \frac{-2\omega_k + 3\delta_k}{\sqrt{13}}, \qquad k = 0, 1, 2. \tag{28}$$

The geometric reason why Y_k is null is simple. For a fixed k, the twist ω_k mixes a time-direction component into e^k, while the mixing δ_k conversely mixes a spatial component into e^0. The Y_k direction is the combination in which these two deformations cancel in the quadratic form of the metric, yielding $G\,Y_k = 0$. However, the coframe deformation itself does not vanish, so this direction is not gauge but remains as a *constraint-like auxiliary.*

In this basis,

$$G\,Y_k = 0, \qquad G_{\rm phys} = 769.8286\,I_3, \qquad H_{\rm phys} = 3223.06\,I_3, \tag{29}$$

and the physical generalised eigenvalue is

$$\lambda_{\rm phys} = \left.\frac{u^T H u}{u^T G u}\right|_{u=(-2,3)} = \frac{41899.77}{10007.77} = 4.1867 > 0. \tag{30}$$

The physical spin-1 triplet is therefore locally stable.

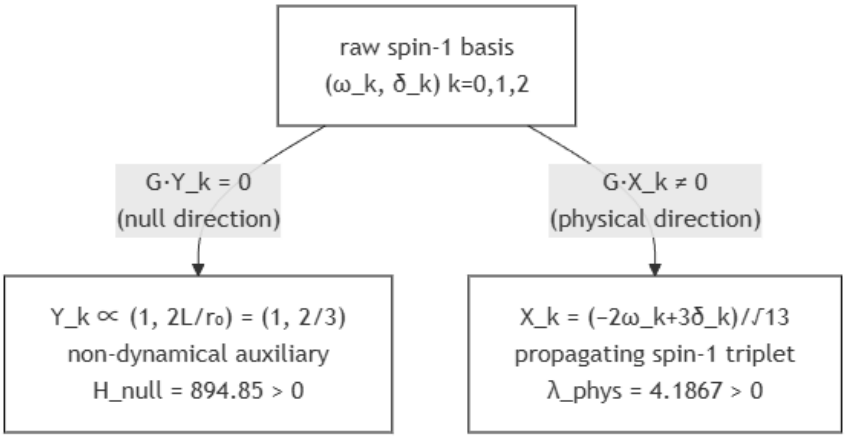


Figure 10: Decomposition from the raw spin-1 basis to the physical/auxiliary basis. The rank-deficiency of G is the origin of the decomposition.

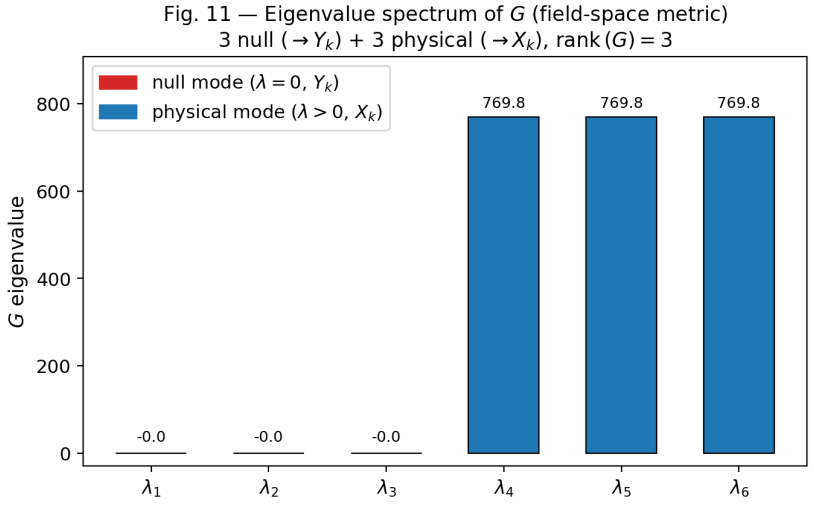


Figure 11: Eigenvalue spectrum of the field-space metric G in the spin-1 sector. Three eigenvalues are exactly zero (null directions Y_k) and three are degenerate at $G_{\rm phys} = 769.83$ (physical directions X_k), reflecting the $SO(3)$ Schur structure.

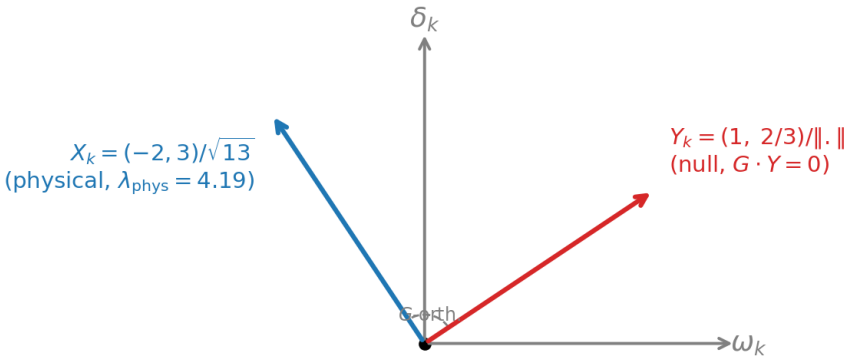


Figure 12: Basis rotation from raw spin-1 variables (ω_k, δ_k) to the physical/auxiliary basis (X_k, Y_k). The null direction $Y_k \propto (1, 2L/r_0)$ is a non-dynamical auxiliary; the orthogonal direction X_k carries the propagating spin-1 degree of freedom.

5.3 Why Y_k Is Not Gauge

It is important not to confuse Y_k with gauge freedom or variable redundancy. For axis $k = 0$, the variation along the null direction gives

$$\delta e^3_{\text{twist}} = +\frac{2L}{r_0} e^0, \qquad \delta e^3_{\text{mixing}} = -\frac{2L}{r_0} e^0, \tag{31}$$

so $\delta e^3 = 0$ in total. However,

$$\delta e^0 = +\frac{2L}{r_0} e^3 \tag{32}$$

survives, producing the metric variation

$$\delta g_{\tau,\sigma^0} = 1 \neq 0. \tag{33}$$

Hence this direction is *not* gauge.

Furthermore, variable redundancy would require:

- $\delta g_{\mu\nu} = 0$ (metric variation vanishes),

- $H_{\text{null}} = 0$ (Hessian along the null direction vanishes), and
- λ_{eff} finite.

In reality,

$$H_{\text{null}} = 894.845 > 0, \qquad \lambda_{\text{eff}} \to +\infty. \tag{34}$$

Therefore Y_k is correctly interpreted as an auxiliary direction without a kinetic term, i.e. a *non-dynamical auxiliary*. Full details of the variable-redundancy exclusion are given in Appendix E.4.

5.4 Quadratic Decoupling

We evaluated whether the physical spin-1 mode mixes with other sectors at quadratic order. The mixed Hessians

$$M_{Ai}^{(qX)} = \frac{\partial^2 V}{\partial q_A \partial X_i}, \qquad M_i^{(rX)} = \frac{\partial^2 V}{\partial r\, \partial X_i}, \qquad M_i^{(\eta X)} = \frac{\partial^2 V}{\partial \eta\, \partial X_i} \tag{35}$$

(15, 3, and 3 components respectively; 21 in total) were evaluated by numerical finite differences, and

$$M_{Ai}^{(qX)} = 0, \qquad M_i^{(rX)} = 0, \qquad M_i^{(\eta X)} = 0 \tag{36}$$

were all confirmed. Step-size stability and the parity check $V(p, +X_k) - V(p, -X_k) = 0$ are detailed in Appendix F. Quadratic decoupling is therefore a structural fact in the final basis, not a numerical artefact.

5.5 Final Quadratic Sector for Spin-1

Summarising, the quadratic contribution of the spin-1 sector is

$$\mathcal{L}_X + \mathcal{L}_Y = \frac{1}{2}\dot{X}^T G_{\text{phys}} \dot{X} - \frac{1}{2} X^T H_{\text{phys}} X - \frac{1}{2} Y^T H_{\text{null}} Y. \tag{37}$$

The field Y has no kinetic term, and the negative direction apparent in the raw basis is not a physical instability. The complete block decomposition of $\mathcal{L}_{\text{quad}}$ including all sectors is presented in §6.

The conclusions of this section are the following three points.

1. The raw basis (ω_i, δ_i) is not the final physical basis.
2. Y_k is neither gauge nor redundancy but a non-dynamical auxiliary.
3. X_k is the unique propagating spin-1 triplet and decouples from all other sectors at quadratic order.

6 Unified Geometric Landau EFT

This section consolidates the results of §§3–5 into a single final effective theory. The key point is that in the final basis the quadratic sector is block-diagonally organised and the allowed interaction terms are described transparently by the invariant basis and representation- theoretic selection rules.

6.1 Raw Quadratic Data

In the raw basis the effective potential is

$$\begin{aligned} V_{\text{eff}} = V_0 + \tfrac{1}{2}\mu_q^2\,\mathrm{tr}(h^2) + \tfrac{1}{2}\mu_\omega^2|\omega|^2 + \tfrac{1}{2}\mu_\delta^2|\delta|^2 + m_{\omega\delta}\,\omega\cdot\delta \\ + c_\omega\,h_{ij}\omega_i\omega_j + c_\delta\,h_{ij}\delta_i\delta_j + O(4), \end{aligned} \tag{38}$$

where

$$\mu_q^2 = 48\pi^2, \quad \mu_\omega^2 = 52.6379, \quad \mu_\delta^2 = 3789.9295, \quad m_{\omega\delta} = -631.6544, \tag{39}$$

$$c_\omega = 105.28, \qquad c_\delta = 1894.96, \qquad c_{\omega\delta} = 0 \quad \text{(at the isotropic point)}. \tag{40}$$

The raw geometric description is necessary for tracing the geometrical origins but is redundant as an organiser of physical degrees of freedom. The final basis is derived via the physical/auxiliary decomposition of §5.

6.2 Final Quadratic EFT

In the final basis the quadratic sector decomposes conceptually as

$$\mathcal{L}_{\text{quad}} = \mathcal{L}_r + \mathcal{L}_{q_A} + \mathcal{L}_X + \mathcal{L}_Y + \mathcal{L}_{\eta,V}, \tag{41}$$

with

$$\mathcal{L}_{q_A} = -\frac{1}{2}\mu_q^2\sum_{A=1}^{5} q_A^2, \qquad \mathcal{L}_X = \frac{1}{2}\dot{X}^T G_{\text{phys}}\dot{X} - \frac{1}{2}X^T H_{\text{phys}} X, \qquad \mathcal{L}_Y = -\frac{1}{2}Y^T H_{\text{null}} Y. \tag{42}$$

Since all 21 mixed Hessian components between X_k and (q_A, r, η) vanish, quadratic decoupling of the spin-1 sector is an exact statement in the final basis. All 10 spin-2–torsion cross-Hessian components also vanish at the isotropic point, and the radial–V cross-Hessian is zero as well. However, the radial–η cross-Hessian is non-zero (≈ 118.4), so the scalar/torsion subsector forms a coupled (r, η) block with V remaining independent. Thus the natural block decomposition at quadratic order is

$$(q_A) \;\oplus\; (X_k) \;\oplus\; (Y_k) \;\oplus\; \Big[(r,\eta)\text{ block} \;\oplus\; (V)\Big]. \tag{43}$$

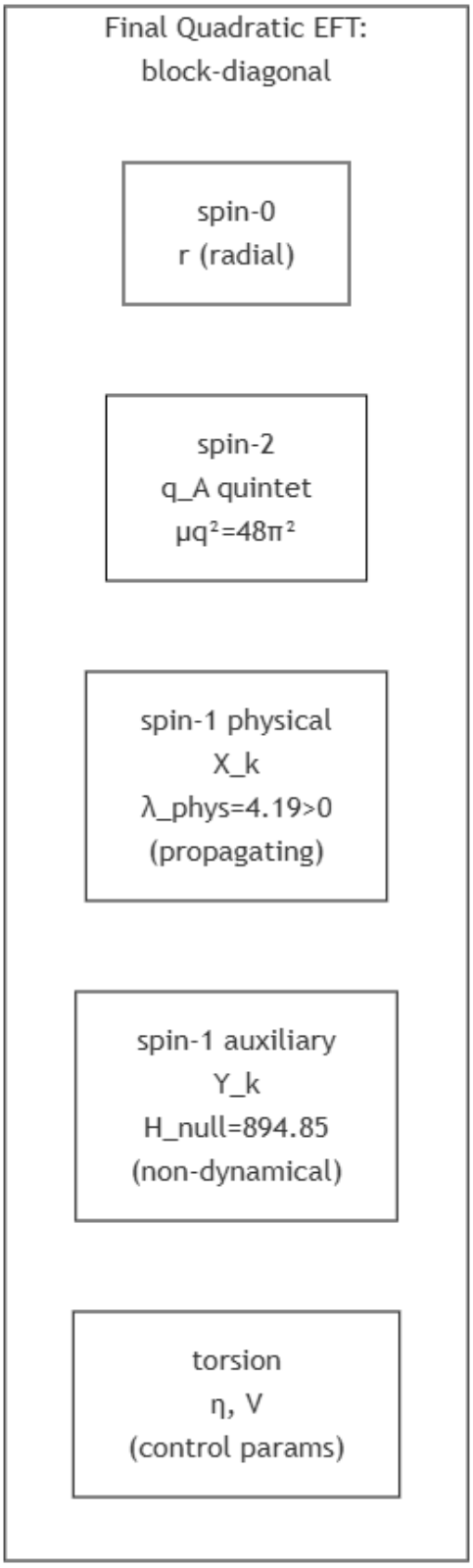


Figure 13: Quadratic block organisation of the final sector. The spin-1 sector fully decouples; the spin-2 sector decouples from torsion; the scalar/torsion subsector contains a coupled (r, η) block.

6.3 Invariant Basis and Selection Rules

The low-order scalars of the Landau-EFT are organised by the invariant basis

$$I_2 = \mathrm{tr}(h^2), \qquad I_3 = h_{ij}\omega_i\omega_j, \qquad I_4 = \mathrm{tr}(h^3), \qquad W_2 = \omega_i\omega_i. \tag{44}$$

An exact check confirms $I_3 = q_A F_A(\omega)$ in the final T_A basis. At quartic order the natural helper invariants are I_2^2, $I_2 W_2$, W_2^2.

This organisation implies:

- Representative allowed cubic terms are $h_{ij}\omega_i\omega_j$ and $h_{ij}\delta_i\delta_j$.
- Quadratic cross-terms in X_k vanish in the final basis.
- Known forbidden terms H_{sym} and K_{sym} are excluded by the representation-theoretic selection rule.

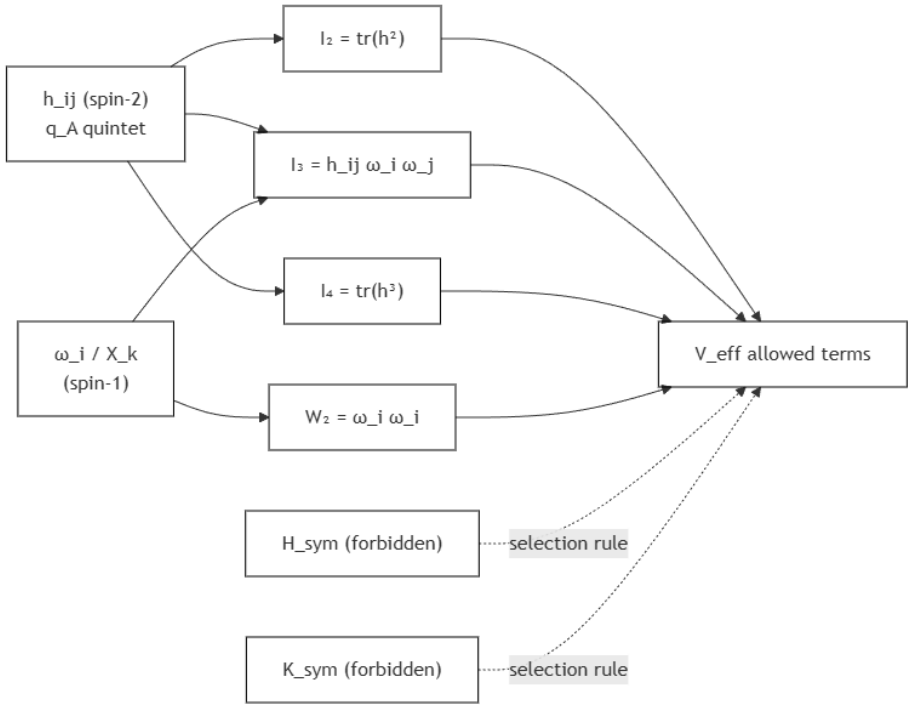


Figure 14: Connection structure of low-order invariants in the Landau-EFT. Solid arrows indicate allowed terms; dashed arrows indicate terms forbidden by selection rules.

6.4 Final Mode Dictionary

We recapitulate the mode dictionary adopted in this paper.

Variable	Final classification	Role	Comment
r	spin-0	isotropic radius	equilibrium $r_0 = 3$
q_A	spin-2	full shear quintet	$\mu_q^2 = 48\pi^2$
X_k	spin-1 physical	propagating mixed mode	quadratic decoupled
Y_k	spin-1 auxiliary	non-propagating auxiliary	not gauge
η	spin-0 torsion	AX parameter	input/control
V	spin-0 torsion	VT parameter	input/control

Table 3: Final mode dictionary. The raw geometric variables (ω_i, δ_i) are an intermediate description; they are translated to X_k and Y_k in the final physical dictionary.

The block decomposition of the quadratic sector is as given in §6.2: the spin-1 and spin-2 sectors decouple, while the scalar/torsion subsector retains the coupled (r, η) block identified there.

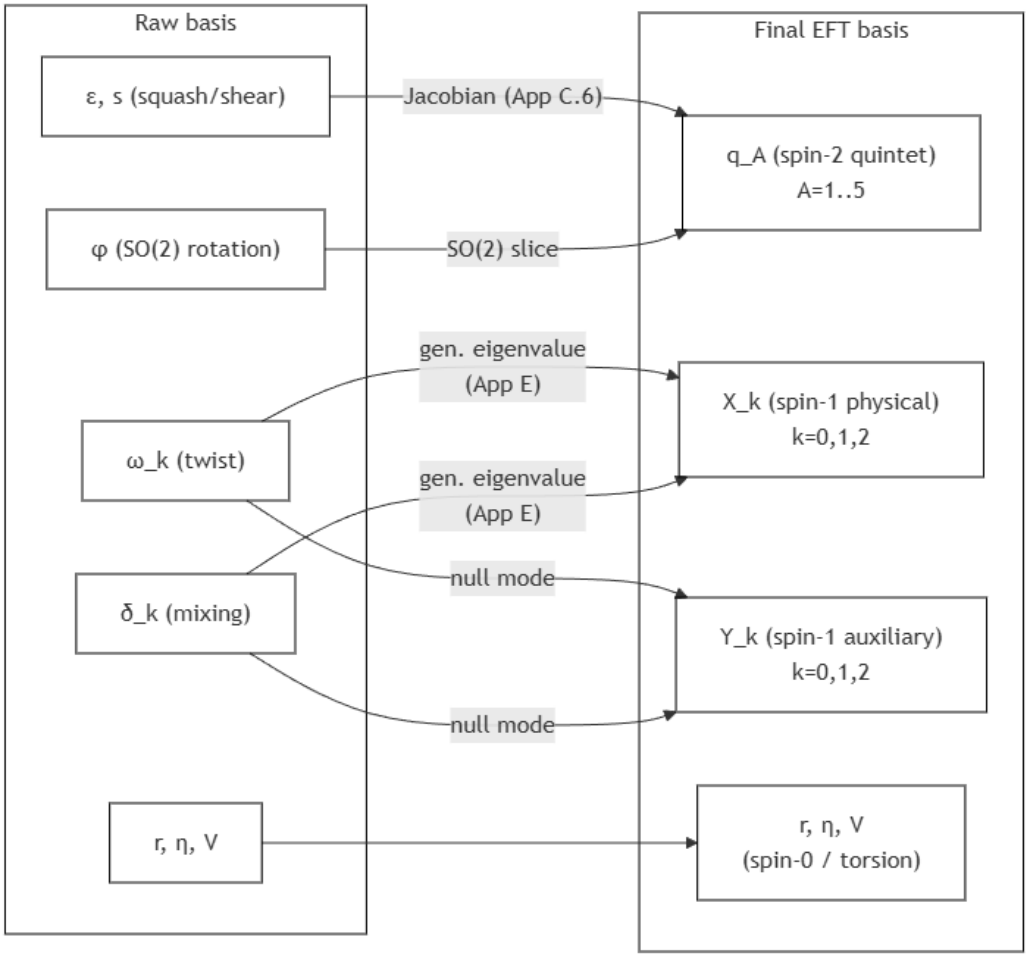


Figure 15: Overall transformation from raw geometric variables to the final EFT basis. Arrows indicate the assignment of each variable.

6.5 Interpretation as a Geometric Landau EFT

From the above results the model can be understood not merely as a description of a numerical potential in the gravitational minisuperspace, but as a finite- degree-of-freedom *geometric Landau-EFT*:

1. Order-parameter-like geometric degrees of freedom are organised by irreducible representations.

2. The quadratic sector is block-organised into representation-theoretic blocks. The spin-1 sector fully decouples; the scalar/torsion side has the refined block structure — a coupled (r, η) block and an independent V direction.

3. Allowed interactions are described by the invariant basis and representation-theoretic selection rules.

This section thus translates the behaviour of the geometric degrees of freedom in the $S^3 \times S^1$ topology of the minisuperspace reduction into a unified theoretical language.

The cross-Hessian analysis of §6.2 shows that this block organisation is exact at quadratic order (see also Appendix H.3 for the full numerical tables). In addition, the torsion block $H^{(\eta,V)}$ at the isotropic point is diagonal and doubly degenerate with positive eigenvalues (≈ 355.3), while the coupled (r, η) 2×2 block becomes saddle-point-type (indefinite) in the AX mode, reflecting the flat-direction nature of r, yielding $\lambda_{\min} < 0$.

7 Physical Interpretation and Outlook

This section clarifies what has been completed and what has not yet been settled.

7.1 What Has Been Completed in homogeneous bulk geometric sector

The principal items established in this paper at the classical quadratic level of the homogeneous minisuperspace geometric sector are summarised in §8. Here we add a remark on the scope limitation.

"Quadratic block organisation" means that the homogeneous bulk sector can be decomposed into spin-2, physical spin-1, auxiliary spin-1, and scalar/torsion subsectors. Note, however, that the scalar/torsion subsector is not fully diagonal in the (r, η, V) variables: at quadratic order the radial mode mixes with axial torsion, while the V direction remains separate at the isotropic point.

It is important to emphasise that *inhomogeneous modes are not addressed here.* This paper concerns the classical EFT structure of the homogeneous bulk geometric sector.

7.2 What Is Not Claimed Here

The following claims are deliberately withheld.

- Full stability including inhomogeneous modes beyond minisuperspace.
- Identification of the generalised eigenvalue $\lambda_{\rm phys}$ with a universal mass squared.
- A claim that the full 4-dimensional deformed geometry constitutes a new Lie group manifold.
- Complete action generation including arbitrary non-invariant polynomials.

These are not in contradiction with the results of this paper; they are consciously outside the stated scope.

7.3 Relation to the Thermal Interpretation of the S^1 Direction

Since paper I [1], the S^1 direction has been regarded as a natural connection point to a thermal interpretation. Although this paper does not explicitly discuss that connection, the final-basis organisation of the homogeneous geometric sector provides a firmer foundation for future discussions of:

- the stable sector of the background geometry;

- control parameters coupled to thermal deformations;
- the separation of bulk geometric response from thermally dependent responses.

7.4 Toward Future Extensions

The future tasks for this model are the following.

1. Introduction of inhomogeneous fluctuations including local degrees of freedom.
2. Settings involving boundaries and interfaces.
3. Connection between homogeneous bulk modes and localised modes.
4. Bridges to physical observables including thermal and Lorentzian interpretations.

8 Conclusion

We have reorganised the homogeneous $S^3 \times S^1$ minisuperspace geometry of Einstein–Cartan + Nieh–Yan theory along the final irreducible basis. As a result, the following five structures close as a single theoretical framework:

- Pontryagin protection in the AX/VT torsion modes,
- completion of the spin-2 quintet by the full shear tensor,
- physical/auxiliary decomposition of the twist-mixing system,
- quadratic decoupling in the final basis, and
- a unified geometric Landau-EFT in the invariant basis.

In particular, the central message of this paper can be distilled into one sentence:

> *The homogeneous geometric sector of the minisuperspace is a completed geometric Landau-EFT, organised along the spin-0/1/2 irreducible representations.*

From this conclusion, the present model establishes one closed framework for the classical homogeneous geometric sector. On the other hand, the next challenges — thermal interpretation, local degrees of freedom, and interface analysis — are left to future work. This paper provides the theoretical junction that delineates those branches.

Acknowledgments

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The AI tools did not determine the scientific claims of this work and were not used to generate or modify research data or evidentiary figures. The author takes full responsibility for the content and for any remaining errors.

This work was developed within the informal collaborative project **DPPU** (**D**onut-like topology, **P**lanck-scale compactness, **P**recession dynamics, **U**niverse).

Availability and Contact

Code and Data Availability The code and supplemental data associated with this work are available at:

- Zenodo: `10.5281/zenodo.19144481`
- GitHub: `https://github.com/Muacca/DPPUv2-paper03`

They are released under the MIT License.

Citation If you use these materials, please cite the associated manuscript and Zenodo record:

> Muacca, "Unified Geometric Landau EFT of Homogeneous $S^3 \times S^1$ Minisuperspace in Einstein–Cartan + Nieh–Yan Theory."

Contact For questions, bug reports, or collaboration inquiries:

- Email: `muacca@dmwp.jp`
- GitHub Issues: `https://github.com/Muacca/DPPUv2-paper03/issues`

A Conventions and Computational Pipeline

This appendix collects the notation conventions and the computational pipeline for reproducing the results of this paper.

A.1 Notation Conventions

- Spatial tensor indices: $i, j = 1, 2, 3$.
- Spin-2 basis label: $A = 1, \ldots, 5$.
- Spin-1 triplet label (consistent with the scripts): $k = 0, 1, 2$.
- Volume factor: $\mathrm{Vol} = 2\pi^2 L r^3$.
- Effective potential: $V_{\mathrm{eff}} = -\left(\frac{R}{2\kappa^2} + \theta_{\mathrm{NY}} N\right) \mathrm{Vol}$.
- Pontryagin density: $P = \langle R, \star R \rangle = \frac{1}{2}\, \varepsilon^{abcd} R_{abef} R_{cd}{}^{ef}$.

A.2 Where the Mode Dictionary Is Fixed

The mode dictionary is presented in a distributed manner.

1. §2 provides the overview from raw geometric variables to the final basis.
2. §5 establishes the physical/auxiliary classification of the spin-1 sector.
3. §6 fixes the definitive dictionary adopted in this paper.
4. Appendix H.3 collects the numerical anchors corresponding to that dictionary.

A.3 Computational Pipeline

The numerical and symbolic computations of this paper are conceptually organised in the following order.

1. Specify the homogeneous frame and the structure constants $C^i{}_{jk}$.
2. Compute the curvature, Ricci scalar, Nieh–Yan density, and Pontryagin density from the Einstein–Cartan connection.
3. Assemble V_{eff} and evaluate the Hessian and generalised eigenvalues at the isotropic point and at deformed points.
4. Organise the full shear basis T_A, spin-1 physical/auxiliary basis (X, Y), and torsion modes into the final basis.

5. As additional consistency checks, independently verify the cross-Hessian, homogeneous consistency, and invariant generation with separate scripts.

A.4 Scripts

The main scripts, organised by purpose, are listed below. The principal appendices (B, D, E, F, G, H) also contain references to the corresponding scripts; see each appendix for details.

Pontryagin protection (Appendix B)

- `proofs/riemann_block_classification.py`
- `proofs/pontryagin_block_decomposition.py`
- `proofs/plucker_cancellation_proof.py`
- `proofs/pfaffian_decomposability.py`
- `proofs/vt_mode_cancellation.py`
- `proofs/analytic_proof_P_equals_zero.py`
- `paper03/pontryagin_scan_s3.py`
- `paper03/pontryagin_scan_nil3.py`
- `paper03/pontryagin_anisotropic_twist.py`
- `paper03/torsion_mode_analysis.py`

Spin-2 sector (Appendix D)

- `paper03/spin2_schur_degeneracy.py`
- `paper03/spin2_spectrum_splitting.py`
- `paper03/spin2_diagonal_hessians.py`
- `paper03/spin2_offdiag_hessian.py`

Spin-1 physical/auxiliary decomposition (Appendix E)

- `paper03/spin1_null_mode_analysis.py`
- `paper03/spin1_kinetic_metric.py`
- `paper03/spin1_auxiliary_exclusion.py`

Cross-Hessian and final-basis consistency (Appendices F, G)

- `paper03/spin1_cross_hessian.py`
- `paper03/homogeneous_consistency.py`
- `paper03/eft_invariant_generation.py`

Scalar/torsion sector and numerical tables (Appendix H)

- `paper03/spin0_torsion_cross_hessian.py`
- `paper03/spin0_reta_block_eigenvalues.py`
- `paper03/spin0_rLetaV_hessian.py`

A.5 Sanity Checks

All conclusions used in the main text have passed at least the following sanity checks.

1. Reproduction of raw spin-1 quadratic coefficients $(\mu_\omega^2, \mu_\delta^2, m_{\omega\delta})$.
2. All 21 cross-Hessians are zero.
3. Homogeneous Maurer–Cartan consistency.
4. Exact check of the invariant basis $\{I_2, I_3, I_4, W_2\}$.

These constitute reproducibility information supporting the main propositions of this paper.

B Detailed Proof of the Pontryagin Protection Theorem

This appendix derives the Pontryagin protection theorem of §3 algebraically.

B.1 Statement

Theorem 2 (P0, restated)**.** *For the AX and VT modes and for the homogeneous minisuperspace topologies under consideration,* $P = \langle R, \star R\rangle \equiv 0$.

B.2 AX Sector: Algebraic Proof of Plücker/Pfaffian Cancellation

Coframe definition

The $S^3 \times S^1$ coframe used throughout this section is

$$e^i = r\, M^i{}_j\, \sigma^j \quad (i = 0, 1, 2), \qquad e^3 = L\Big(d\tau + \omega_k\, \sigma^k\Big)\,, \tag{45}$$

where σ^i are the left-invariant 1-forms of S^3 and $M^i{}_j = (\exp h)^i{}_j$ with $h = \sum_A q_A T_A$. For isotropic AX mode we set $h = 0$ ($M = I$) and $\omega_k = \omega$ (all components equal).

The structure constants from the Maurer–Cartan equations $de^k = (1/r)\varepsilon_{ijk}\, e^i \wedge e^j$, $de^3 = (2L\omega/r^2)\varepsilon_{ij}\, e^i \wedge e^j$ are

$$C^k_{ij} = \frac{2}{r}\varepsilon_{ijk}, \qquad C^3_{ij} = \frac{4L\omega}{r^2}\varepsilon_{ij}. \tag{46}$$

Defining

$$c \equiv \frac{1+\eta}{r}, \qquad f \equiv \frac{2L\omega}{r^2}, \tag{47}$$

the non-zero EC connection components are

$$\omega^k_{ij} = c\,\varepsilon_{ijk}, \qquad \omega^3_{ij} = f\,\varepsilon_{ij}, \qquad \omega^i_{3j} = -f\,\varepsilon_{ij}, \qquad \omega^i_{j3} = +f\,\varepsilon_{ij}. \tag{48}$$

Riemann tensor formula for homogeneous spaces

For homogeneous spaces the frame derivatives vanish and the Riemann tensor is

$$R^a{}_{bcd} = \omega^a{}_{ec}\,\omega^e{}_{bd} - \omega^a{}_{ed}\,\omega^e{}_{bc} + \omega^a{}_{be}\,C^e{}_{cd}. \tag{49}$$

Representative components for $(c, d) = (0, 1)$

The key Riemann components needed for the Plücker condition are

$$R_{0101} = \frac{1-\eta^2}{r^2} + f^2, \tag{50}$$

$$R_{2301} = 0, \tag{51}$$

$$R_{0201} = -f^2, \tag{52}$$

$$R_{1301} = \frac{f(\eta-1)}{r}, \tag{53}$$

$$R_{0301} = \frac{f(1-\eta)}{r} = -R_{1301}, \tag{54}$$

$$R_{1201} = f^2. \tag{55}$$

Verification: Pfaffian $= 0$ for $(c, d) = (0, 1)$

$$\begin{aligned} \mathrm{Pf}\,\omega^{(01)} &= R_{0101}\cdot R_{2301} - R_{0201}\cdot R_{1301} + R_{0301}\cdot R_{1201} \\ &= \left(\tfrac{1-\eta^2}{r^2} + f^2\right)\cdot 0 - (-f^2)\cdot \tfrac{f(\eta-1)}{r} + \tfrac{f(1-\eta)}{r}\cdot f^2 \\ &= \frac{f^3(\eta-1)}{r} + \frac{f^3(1-\eta)}{r} = 0. \qquad \checkmark \end{aligned} \tag{56}$$

Verification for $(c, d) = (0, 3)$ (S3-type)

Using $C^e_{03} = 0$,

$$R_{0103} = -fc,\ R_{2303} = f^2,\ R_{0203} = -fc,\ R_{1303} = f^2,\ R_{0303} = -f^2,\ R_{1203} = 0. \tag{57}$$

$$\mathrm{Pf}\,\omega^{(03)} = (-fc)\cdot f^2 - (-fc)\cdot f^2 + (-f^2)\cdot 0 = -f^3c + f^3c = 0. \qquad \checkmark \tag{58}$$

Generalisation to all (c, d) pairs

The six (c, d) pairs fall into the SS-type $\{01, 02, 12\}$ and the S3-type $\{03, 13, 23\}$.

- **SS-type**: The cyclic symmetry of $\{0, 1, 2\}$ (the $SO(3)$ symmetry of S^3) gives the same algebraic structure as $(0, 1)$. The Plücker condition holds for the same algebraic reason.
- **S3-type**: As seen for $(0, 3)$, the key is $C^e_{i3} = 0$ (the τ-direction generates no structure constants). This holds for all S3-type pairs.
- P_{33} **terms**: $R_{3\ldots3\ldots}$ vanishes trivially by antisymmetry.

Hence the Pontryagin density is established algebraically:

$$P = 2\sum_{(c,d)} \mathrm{Pf}\,\omega^{(cd)} = 0. \tag{59}$$

Remark. The above proof is for isotropic twist. For general vector twist $(\omega_0, \omega_1, \omega_2)$ the same structure of structure constants is preserved, and together with the identity $V_{\text{eff}} = f(I_2)$, it is confirmed symbolically that $P \equiv 0$ (including $\text{Nil}^3 \times S^1$).

B.3 VT Sector: Pfaffian Cancellation

In the VT mode the cancellation mechanism differs. For $S^3 \times S^1$,

$$\sum_{\text{SS}} \text{Pf}\, \omega^{(cd)} = -\frac{16L^2 V\, I_2}{3r^5}, \qquad \sum_{\text{S3}} \text{Pf}\, \omega^{(cd)} = +\frac{16L^2 V\, I_2}{3r^5}, \tag{60}$$

giving $P = \sum_{\text{SS}} \text{Pf} + \sum_{\text{S3}} \text{Pf} = 0$.

The cancellation is not trivial decomposability of each block (as in AX mode), but an exact sign cancellation between the SS and S3 group Pfaffian totals. The sign reversal originates from V entering as pure vector-trace torsion only in the τ direction, so the SS and S3 dual structures act complementarily. The same cancellation holds algebraically for $\text{Nil}^3 \times S^1$ as well.

B.4 MX Sector: the Unique Exception

In the MX mode $\eta \neq 0$ and $V \neq 0$ simultaneously, and the protection breaks. For $S^3 \times S^1$:

$$P(\boldsymbol{\omega}) = \frac{2V\eta\,(V^2 r^2 + 9\eta^2 - 36)}{9r^3} + \frac{40V\eta L^2}{3r^5}\, I_2. \tag{61}$$

Even at the isotropic point ($I_2 = 0$), $P_0 = 2V\eta(V^2 r^2 + 9\eta^2 - 36)/(9r^3) \neq 0$, so the origin of $P \neq 0$ is torsion mixing $V\eta$, not the twist ω.

B.5 NyVariant Independence

The Riemann tensor (and hence P) is independent of the NyVariant (FULL/TT/REE) choice, as guaranteed theoretically by the pipeline computation order. The results by mode are:

Mode	FULL	TT	REE
AX	0	0	0
VT	0	0	0
MX	nonzero	nonzero	nonzero

Table 4: Pontryagin density by torsion mode and NyVariant.

B.6 Interpretation

The key points of this appendix are the following.

1. AX and VT are protected by distinct algebraic mechanisms: AX by individual cancellation per (c, d) block (Plücker type); VT by sign cancellation between SS and S3 group Pfaffians.

2. The MX mode is the unique exception; the twist itself is not the origin.

3. The Pontryagin structure is cleanly separated sector by sector in a manner consistent with the final-basis organisation.

B.7 Computational Verification

The algebraic computations of this appendix are confirmed symbolically by the following scripts.

- AX mode, Riemann block classification, per-(c, d) Pfaffian zero check: `paper03/pontryagin_anisotropic_twist.py`
- VT/MX modes, NyVariant cross-check (all 18 cases): `paper03/torsion_mode_analysis.py`
- Isotropic twist $P \equiv 0$ scan (r, η dependence): `paper03/pontryagin_scan_s3.py`, `paper03/pontryagin_scan_nil3.py`

C Full Shear Tensor Basis and Embedding of (ε, s, ϕ)

This appendix summarises the final basis of the spin-2 sector and its relation to the raw geometric variables (ε, s, ϕ).

C.1 Traceless Symmetric Basis

The full shear tensor is expanded as

$$h_{ij} = \sum_{A=1}^{5} q_A \, (T_A)_{ij}. \tag{62}$$

The explicit matrix representations of $T_1, \ldots, T_5$ are given in §2.3. The following properties hold exactly:

$$\mathrm{tr}(T_A T_B) = \delta_{AB}, \qquad \mathrm{tr}(h) = 0, \qquad \mathrm{tr}(h^2) = \sum_{A=1}^{5} q_A^2. \tag{63}$$

C.2 Embedding of the Old Variables

The raw geometric variables (ε, s, ϕ) provide a description of a particular three-dimensional slice within the spin-2 quintet. In this paper we do not place these slice variables in the primary role; instead we adopt the final basis q_A as the principal variables. Accordingly:

- ε is primarily used for axisymmetric deformations in the $m = 0$ singlet direction;
- s and ϕ are auxiliary parameters for tracking the negative-side condensed phase.

All representation-theoretic statements are always rewritten in q_A.

C.3 Invariants in the Final Basis

It is confirmed exactly that the low-order invariants in the final basis are

$$I_2 = \mathrm{tr}(h^2) = \sum_{A=1}^{5} q_A^2, \qquad I_3 = h_{ij}\omega_i\omega_j, \qquad I_4 = \mathrm{tr}(h^3). \tag{64}$$

C.4 Constant-Shape Homogeneous Consistency

The homogeneous consistency for the constant frame deformation $e^i = r\, M^i{}_j\, \sigma^j$ with the structure-constant transformation

$$C'^i{}_{jk} = M^i{}_a \, (M^{-1})^b{}_j \, (M^{-1})^c{}_k \, C^a{}_{bc} \tag{65}$$

was verified for two representative all-five-nonzero full-shear samples on both S^3 and Nil^3:

- antisymmetry residual = 0,
- Jacobi residual = 0,

both holding exactly. The final spatial ansatz is therefore self-consistent as a homogeneous minisuperspace frame.

C.5 Role of This Appendix

The purpose of this appendix is to clarify the change of viewpoint from raw geometric variables to the final basis. All final statements about spin-2 in this paper are written in q_A and its irreducible decomposition; (ε, s, ϕ) are used only as auxiliary parameters for the phase diagram.

C.6 Jacobian: Old Variables to Final Basis

We give the correspondence between the raw geometric variables (ε, s) and the final basis coordinates (q_1, q_2) explicitly.

For ε—which embeds a T_1-direction deformation exponentially:

$$\varepsilon = \exp\left(\frac{\sqrt{6}}{2}\, q_1\right) - 1, \qquad q_1 = \frac{2}{\sqrt{6}} \ln(1+\varepsilon), \qquad \left.\frac{\partial q_1}{\partial \varepsilon}\right|_{\varepsilon=0} = \frac{2}{\sqrt{6}} = \sqrt{\frac{2}{3}}. \tag{66}$$

For s (deformation in the T_2 direction):

$$s = \exp\left(\frac{q_2}{\sqrt{2}}\right) - 1, \qquad q_2 = \sqrt{2} \ln(1+s), \qquad \left.\frac{\partial q_2}{\partial s}\right|_{s=0} = \sqrt{2}. \tag{67}$$

The angle ϕ appearing in the $s \neq 0$ phase corresponds to an $SO(2)$ rotation in the (q_2, q_3) plane:

$$q_2 = s_q \cos\phi, \qquad q_3 = s_q \sin\phi, \qquad s_q = \sqrt{q_2^2 + q_3^2}. \tag{68}$$

The analogous embedding for the T_4, T_5 directions (off-diagonal components) is structurally identical by symmetry.

The combined Jacobian diagonal at the isotropic point is

$$J = \mathrm{diag}\left(\sqrt{\tfrac{2}{3}},\, \sqrt{2}\right), \tag{69}$$

and the Hessian basis transformation is

$$H_q = J^{-T} H_{\varepsilon s}\, J^{-1}. \tag{70}$$

At the isotropic point, $H_{\varepsilon s} = \mu_q^2\, I_2$ and $J^{-T} J^{-1} \propto I_2$, so Schur degeneracy $H_q = \mu_q^2\, \delta_{AB}$ is preserved under the change of basis.

D Mass Splitting Formulae in the Spin-2 Sector

This appendix collects the residual $SO(2)$ splitting, the closed-form Hessians for all spin-2 modes, and the phase information of the spin-2 sector.

D.1 Residual $SO(2)$ Decomposition

Under a positive-ε squash background, $SO(3) \to SO(2)$ and the spin-2 quintet splits as follows.

Mode	Basis	Residual charge
Singlet	T_1	$m = 0$
Doublet	(T_2, T_3)	$m = \pm 2$
Doublet	(T_4, T_5)	$m = \pm 1$

Table 5: Residual $SO(2)$ decomposition of the spin-2 quintet.

D.2 Isotropic Point

At the isotropic point Schur's lemma gives

$$H_{AB} = 2\lambda\, \delta_{AB}, \qquad \mu_q^2 = 48\pi^2, \tag{71}$$

with all five components fully degenerate.

D.3 Local Splitting around the Isotropic Point

On the axisymmetric slice $s = 0$ the three representative branches are tracked as

$$\mu^2_{m=0} \equiv \mu^2_\varepsilon, \qquad \mu^2_{m=\pm 2} \equiv \mu^2_s, \qquad \mu^2_{m=\pm 1} \equiv \mu^2_\delta. \tag{72}$$

From a numerical scan near the isotropic point, the local linear splittings are approximately

$$\mu^2_{m=0}(\varepsilon) \approx 48\pi^2 - 1.37 \times 10^3\, \varepsilon + O(\varepsilon^2), \tag{73}$$

$$\mu^2_{m=\pm 2}(\varepsilon) \approx 48\pi^2 + 7.95 \times 10^2\, \varepsilon + O(\varepsilon^2), \tag{74}$$

$$\mu^2_{m=\pm 1}(\varepsilon) \approx 48\pi^2 - 3.0 \times 10^2\, \varepsilon + O(\varepsilon^2). \tag{75}$$

The coefficients are local slopes obtained from numerical data at $|\varepsilon| \lesssim 0.1$ and serve as a practical summary of the branch ordering near the isotropic point, not as exact closed-form results.

The direction of splitting is:

- For positive ε, the $m = 0$ singlet softens most rapidly.
- The $m = \pm 2$ doublet hardens in the same region and plays no role in the positive-side instability.
- The $m = \pm 1$ doublet lies between them and remains massive near the critical point.

This is consistent with the description in §4.

D.4 Closed-Form Hessians for All Spin-2 Modes

The following closed-form expressions for H^q_{ii} of the three representative modes use the parameter $a = (1+\varepsilon)^{1/3}$ (physical parameters: $\eta = L = \kappa = 1$, $\alpha = \theta_{\rm NY} = 0$, MX mode).

T_1 **($m = 0$ squash singlet):**

$$H^q_{11}(a, r) = \frac{-160\pi^2 a^6 r + 352\pi^2 r}{3a^8}, \qquad a = (1+\varepsilon)^{1/3}. \tag{76}$$

T_2 **($m = \pm 2$ shear doublet):**

$$H^q_{22}(a, r) = \frac{128\pi^2 a^6 r - 64\pi^2 r}{a^2}, \qquad a = (1+\varepsilon)^{1/3}. \tag{77}$$

T_4 **($m = \pm 1$ off-diagonal doublet):**

$$H^q_{44}(f, r) = \frac{32\pi^2 f^{12} r - 48\pi^2 f^6 r + 32\pi^2 f^3 r + 48\pi^2 r}{f^8}, \qquad f = (1+\varepsilon)^{2/3} = a^2. \tag{78}$$

At the isotropic point $\varepsilon = 0$ ($a = f = 1$), Schur's lemma gives

$$H^q_{11} = H^q_{22} = H^q_{44} = 64\pi^2 r_0, \tag{79}$$

consistent with §4 and confirmed numerically by the scripts.

The exact critical points (both independent of r_0, derived from the closed-form expressions):

$$H^q_{22} = 0 \iff 2a^6 = 1 \Rightarrow \varepsilon_{c-} = \frac{1}{\sqrt{2}} - 1 \approx -0.292893 \quad \text{(exact)}, \tag{80}$$

$$H^q_{11} = 0 \iff 5a^6 = 11 \Rightarrow \varepsilon_{c+} = \sqrt{\frac{11}{5}} - 1 \approx 0.48324 \quad \text{(exact)}. \tag{81}$$

Script references: `paper03/spin2_diagonal_hessians.py` (H_{11}, H_{22} closed forms); `paper03/spin2_offdiag_hessian.py` (H_{44} closed form; Method 2 rationalisation).

D.5 Representative Spectrum Values

ε	$\mu^2_{m=0}$	$\mu^2_{m=\pm2}$	$\mu^2_{m=\pm1}$
0.00	473.74	473.74	473.74
0.10	336.83	589.55	442.41
0.20	219.78	685.05	413.32
0.40	44.91	842.19	359.80
0.482	0.49	891.77	341.58

Table 6: Representative spin-2 branch values on the axiymmetric slice $s = 0$. The soft branch is exclusively $m = 0$; the $m = \pm 2$ branch moves separately toward the negative-side transition.

D.6 Positive Side

On the positive side the T_1 singlet becomes soft first, at

$$\varepsilon_{c+} = \sqrt{\tfrac{11}{5}} - 1 \approx 0.48324 \quad \text{(exact)}, \qquad r_0(\varepsilon_{c+}) \approx 2.8114, \tag{82}$$

where $\mu^2_\varepsilon \to 0$. This threshold is exact and independent of r_0; it is a softening threshold, not the establishment of a fully condensed phase.

D.7 Negative Side

On the negative side the $m = \pm 2$ sector drives the transition to the $s \neq 0$ phase. From the closed-form expression the exact critical point is

$$\varepsilon_{c-} = \tfrac{1}{\sqrt{2}} - 1 \approx -0.292893 \quad \text{(exact)}, \tag{83}$$

consistent with the three-indicator numerical estimate -0.29309 ± 0.00019. The fourth coefficient in the Landau expansion satisfies $b(\varepsilon) \in [2288, 7805] > 0$, confirming a second-order transition.

D.8 Order Parameter Fit

The order parameter is fitted as

$$s_0 \propto |\varepsilon - \varepsilon_c|^\beta, \qquad \beta = 0.4678, \quad R^2 = 0.987, \tag{84}$$

consistent with a continuous transition. β falls slightly below the mean-field value $1/2$ by $1/2 - \beta \approx 0.032$; this paper does not claim to extract the universality class from this minisuperspace EFT.

D.9 Interpretation

The final picture of the spin-2 sector should be understood as mass splitting organised by residual representation, not by the single squash variable. This paper employs representation-theoretic language as the primary description.

D.10 Computational Verification

- Isotropic point Schur degeneracy ($\mu_q^2 = 48\pi^2$): `paper03/spin2_schur_degeneracy.py`
- $SO(2)$ splitting, softening, phase transition (three-indicator analysis, $\varepsilon_{c\pm}$ estimation): `paper03/spin2_spectrum_splitting.py`
- H_{11}^q, H_{22}^q closed forms ($\varepsilon_{c\pm}$ exact): `paper03/spin2_diagonal_hessians.py`
- H_{44}^q closed form (Method 2 rationalisation): `paper03/spin2_offdiag_hessian.py`

E Spin-1 Sector: H, G, and Generalised Eigenvalue Analysis

This appendix summarises the raw quadratic form and the transformation to the final basis for the spin-1 sector.

E.1 Raw Quadratic Blocks

At the isotropic point $r_0 = 3$, $L = 1$,

$$H_2 = \begin{pmatrix} 52.6379 & -631.6544 \\ -631.6544 & 3789.9295 \end{pmatrix}, \qquad G_2 = \begin{pmatrix} 236.870435 & -355.305547 \\ -355.305547 & 532.958162 \end{pmatrix}. \tag{85}$$

The raw Hessian alone is insufficient for determining physical (in)stability; the full pair (H, G) must be treated as a generalised eigenvalue problem.

E.2 Null Mode and Physical Mode

The null direction is

$$Y_k \propto (\omega_k, \delta_k) = \left(1, \frac{2}{3}\right), \qquad k = 0, 1, 2, \tag{86}$$

and the physical direction is

$$X_k = \frac{-2\omega_k + 3\delta_k}{\sqrt{13}}. \tag{87}$$

In this basis,

$$G\,Y_k = 0, \qquad G_{\text{phys}} = 769.8286\, I_3, \qquad H_{\text{phys}} = 3223.06\, I_3, \qquad \lambda_{\text{phys}} = 4.1867 > 0. \tag{88}$$

E.3 Auxiliary Interpretation Summary

Y_k is neither gauge nor variable redundancy. Indeed,

$$\delta g_{\tau,\sigma^0} \neq 0, \qquad H_{\text{null}} = 894.845 > 0, \qquad \lambda_{\text{eff}} \to +\infty, \tag{89}$$

so Y_k is interpreted as a non-dynamical auxiliary without a kinetic term.

E.4 Detailed Exclusion of Variable Redundancy

We detail the exclusion of Y_k as variable redundancy (elaborating §5.3).

Necessary conditions for variable redundancy:

1. $\delta g_{\mu\nu} = 0$ (metric variation vanishes),

2. $H_{\text{null}} = 0$ (Hessian along the null direction vanishes),

3. λ_{eff} finite.

None of these are satisfied, as we now show.

Exclusion of condition 1 ($\delta g_{\tau,\sigma^0} \neq 0$): For axis $k = 0$, the variation along Y_0 direction gives

$$\delta e^3_{\text{twist}} = +\frac{2L}{r_0} e^0, \qquad \delta e^3_{\text{mixing}} = -\frac{2L}{r_0} e^0, \tag{90}$$

so $\delta e^3 = 0$. However,

$$\delta e^0 = +\frac{2L}{r_0} e^3 \tag{91}$$

remains, yielding

$$\delta g_{\tau,\sigma^0} = (\delta e^0)_\tau \, (e^0)_{\sigma^0} = \frac{2L}{r_0} \, (e^3)_\tau \, (e^0)_{\sigma^0} \neq 0. \tag{92}$$

At $r_0 = 3$, $L = 1$ this gives $2L/r_0 = 2/3 \neq 0$. Condition 1 is excluded. The same conclusion holds for $k = 1, 2$.

Exclusion of condition 2 ($H_{\text{null}} > 0$): The Hessian along the null direction $Y = (1,\, 2/3)^T$ is

$$H_{\text{null}} = Y^T H_2 \, Y = 894.845 > 0. \tag{93}$$

Condition 2 is excluded.

Exclusion of condition 3 ($\lambda_{\text{eff}} \to +\infty$): The Rayleigh quotient at $v = Y_k$ gives denominator $G_2 Y_k = 0$, while the numerator $H_{\text{null}} = 894.845 > 0$, so

$$\lambda_{\text{eff}}(Y_k) = \frac{Y^T H_2 Y}{Y^T G_2 Y} \to +\infty. \tag{94}$$

Condition 3 is excluded.

All three necessary conditions for variable redundancy are excluded; Y_k is confirmed to be a **non-dynamical auxiliary**.

E.5 Final Statement

The final conclusions for the spin-1 sector are the following two points.

1. The propagating triplet is X_k alone.

2. Y_k is a constraint-like auxiliary direction and is not included in the physical mass spectrum.

The negative eigenvalue of the raw basis cannot be read directly as a physical instability.

E.6 Computational Verification

- Numerical computation of H_2, G_2 and physical/null basis: `paper03/spin1_null_mode_analysis.py`
- Strict computation of the G metric (field-space metric): `paper03/spin1_kinetic_metric.py`
- Numerical confirmation of variable redundancy exclusion ($\delta g \neq 0$, $H_{\text{null}} > 0$, $\lambda_{\text{eff}} \to \infty$): `paper03/spin1_auxiliary_exclusion.py`

F Cross-Hessian Checks and Quadratic Decoupling

This appendix demonstrates that the physical spin-1 mode X_k decouples from all other sectors at quadratic order.

F.1 Setup

The physical basis is

$$X_k = \frac{-2\omega_k + 3\delta_k}{\sqrt{13}}. \tag{95}$$

Each mixed Hessian is evaluated by a centred finite difference:

$$\frac{\partial^2 V_{\text{eff}}}{\partial p\, \partial X_k} \approx \frac{V(p_0 + h, +X_k) - V(p_0 + h, -X_k) - V(p_0 - h, +X_k) + V(p_0 - h, -X_k)}{4h^2}. \tag{96}$$

F.2 Spin-1 × Spin-2

$$M_{Ai}^{(qX)} = \left.\frac{\partial^2 V_{\text{eff}}}{\partial q_A \partial X_i}\right|_{\text{iso}} = 0 \tag{97}$$

holds for all 15 components.

Variable	X_0	X_1	X_2
ε	0.0000	0.0000	0.0000
s	0.0000	0.0000	0.0000
q_3	0.0000	0.0000	0.0000
q_4	0.0000	0.0000	0.0000
q_5	0.0000	0.0000	0.0000

Table 7: Spin-1 × spin-2 cross-Hessian: all 15 components vanish.

F.3 Spin-1 × Radial and Torsion

$$M_i^{(rX)} = 0, \qquad M_i^{(\eta X)} = 0 \tag{98}$$

both hold.

Sector	X_0	X_1	X_2
r	0.0000	0.0000	0.0000
η	0.0000	0.0000	0.0000

Table 8: Spin-1 × radial and torsion cross-Hessians: all 6 components vanish.

F.4 Step-Size Stability

The zero determination remains stable across step sizes.

h	Max absolute value over all 21 components
10^{-1}	0.0
10^{-2}	0.0
10^{-3}	0.0
5×10^{-4}	0.0
10^{-4}	0.0

Table 9: Step-size stability of the cross-Hessian zero determination.

F.5 Evenness under $X_k \to -X_k$

Even for representative non-zero backgrounds,

$$V(p, +X_k) - V(p, -X_k) = 0 \tag{99}$$

holds numerically. Thus, at least in the current implementation, V_{eff} behaves as an even function of X_k under sign reversal, and the cross-Hessian consequently vanishes.

F.6 Scope Boundary

What this result asserts is *quadratic decoupling* in the final basis. It does not immediately imply full decoupling at nonlinear order or for arbitrary backgrounds. This paper states the block-diagonal EFT of §6 with this scope explicitly declared.

F.7 Computational Verification

- All 21 component finite-difference evaluations (including step-size stability and parity checks): `paper03/spin1_cross_hessian.py`
- Homogeneous consistency (antisymmetry and Jacobi pass; linked to Appendix C): `paper03/homogeneous_consistency.py`
- Exact check of the invariant basis (linked to Appendix G): `paper03/eft_invariant_generation.py`

G Landau-EFT Selection Rules and Forbidden Terms

This appendix briefly summarises the final invariant basis adopted in paper III and the selection rules on it.

G.1 Final Invariant Basis

The low-order invariants adopted are

$$I_2 = \mathrm{tr}(h^2), \quad I_3 = h_{ij}\omega_i\omega_j, \quad I_4 = \mathrm{tr}(h^3), \quad W_2 = \omega_i\omega_i. \tag{100}$$

At quartic order the auxiliary invariants

$$I_2^2, \qquad I_2 W_2, \qquad W_2^2 \tag{101}$$

are employed.

G.2 Explicit Form of I_3

In the final T_A conventions,

$$\begin{aligned} I_3 = {} & \frac{q_1}{\sqrt{6}}\left(\omega_1^2 + \omega_2^2 - 2\omega_3^2\right) + \frac{q_2}{\sqrt{2}}\left(\omega_1^2 - \omega_2^2\right) + \sqrt{2}\, q_3\omega_1\omega_2 \\ & + \sqrt{2}\, q_4\omega_1\omega_3 + \sqrt{2}\, q_5\omega_2\omega_3. \end{aligned} \tag{102}$$

It is confirmed exactly that this gives $I_3 = q_A F_A(\omega)$.

G.3 Allowed and Forbidden Structures

In the final basis the spin-2 $\times$ spin-1 cubic coupling is determined by the Clebsch–Gordan decomposition

$$\text{spin-2} \otimes \text{spin-1} \otimes \text{spin-1}. \tag{103}$$

Since $\omega_i\omega_j$ and $\delta_i\delta_j$ are symmetric products of commuting variables,

$$\mathrm{Sym}^2(\text{spin-1}) = \text{spin-0} \oplus \text{spin-2}, \tag{104}$$

and no antisymmetric spin-1 component appears. Therefore the independent cubic couplings that produce a scalar by contracting with the spin-2 tensor h_{ij} reduce to their spin-2 projections:

$$h_{ij}\omega_i\omega_j, \qquad h_{ij}\delta_i\delta_j. \tag{105}$$

In the language of the final basis:

- $h_{ij}\omega_i\omega_j$ and $h_{ij}\delta_i\delta_j$ are representative allowed cubic terms.
- Quadratic cross-terms in X_k vanish in the final basis.
- Known forbidden terms H_{sym} and K_{sym} are excluded because they cannot be reduced to a symmetric-rank-2 spin-2 projection; they do not satisfy the requisite condition and are therefore not adopted as independent invariants in the final EFT.

G.4 Boundary of the Claim

What this appendix claims is the low-order invariant organisation for the final basis adopted in this paper. The following are *not* claimed:

- Complete generation of arbitrary non-invariant polynomials.
- Unique classification of all nonlinear couplings.

This paper uses the label "unified Landau-EFT" with this limitation explicitly stated.

G.5 Computational Verification

- Invariant generation and exact assertion pass: `paper03/eft_invariant_generation.py`

H Numerical Tables and Supplemental Data

This appendix tabulates the principal numerical values used throughout paper III.

H.1 Core Numerical Dictionary

Quantity	Value	Interpretation
r_0	3.0	isotropic-point radius
μ_q^2	$48\pi^2 \approx 473.74$	spin-2 quintet mass
ε_{c+}	0.48324	positive-side softening threshold
ε_{c-}	-0.29309 ± 0.00019	negative-side second-order transition
μ_ω^2	52.6379	raw twist Hessian
μ_δ^2	3789.9295	raw mixing Hessian
$m_{\omega\delta}$	-631.6544	raw bilinear mixing
$G_{\omega\omega}$	236.870435	raw kinetic metric
$G_{\delta\delta}$	532.958162	raw kinetic metric
$G_{\omega\delta}$	-355.305547	raw kinetic metric
H_{phys}	3223.06	physical spin-1 Hessian
G_{phys}	769.8286	physical spin-1 kinetic metric
λ_{phys}	4.1867	physical generalised eigenvalue
H_{null}	894.845	auxiliary curvature
c_ω	105.28	$h\omega\omega$ cubic coupling
c_δ	1894.96	$h\delta\delta$ cubic coupling

Table 10: Core numerical dictionary for paper III.

H.2 Cross-Hessian Summary

Block	Components	Result
Spin-1 $\times$ spin-2	15	all zero
Spin-1 $\times$ radial	3	all zero
Spin-1 $\times$ torsion	3	all zero
Spin-2 $\times$ torsion (η and V)	10	all zero
Radial $\times$ V	1	all zero
Radial $\times$ η	1	nonzero (≈ 118.4)

Table 11: Cross-Hessian summary at the isotropic point. The spin-1 sector fully decouples; the spin-2 sector decouples from torsion; the scalar/torsion subsector contains a coupled (r, η) block.

H.2A. Supplemental Scalar-Modulus Check with L Released

In the main text the S^1 scale L is fixed at $L = 1$. Here we temporarily release L and record supplemental results from a numerical evaluation of the 4×4 Hessian $H^{(r,L,\eta,V)}$ near the isotropic point.

Conditions: $r_0 = 3$, $L_0 = 1$, $\kappa = 1$, $\alpha = \theta_{\mathrm{NY}} = 0$; centred difference with $h = 10^{-3}$ (stability confirmed at multiple step sizes).

Background	H_{LL}	H_{rL}	$H_{L\eta}$	H_{LV}	$\lambda_{\min}$
AX	≈ 0	-177.65	355.31	0	-349.79
VT	≈ 0	-59.22	0	355.31	-360.20
MX	≈ 0	≈ 0	355.31	355.31	-437.45

Table 12: 4×4 Hessian $H^{(r,L,\eta,V)}$ with L released. All cases are indefinite (saddle-point type).

The full 4×4 eigenvalues are:

- AX: $-349.8,\ 130.1,\ 355.3,\ 575.0$
- VT: $-360.2,\ 121.4,\ 355.3,\ 712.6$
- MX: $-437.5,\ 23.7,\ 412.0,\ 830.9$

Since $H_{LL} \approx 0$, L is a flat direction in the quadratic approximation at the isotropic point, analogous to r. The $L = 1$ normalisation in the main text is therefore appropriate. Rather than a standard scalar degree of freedom with a stable mass, L is a modulus-type auxiliary variable that possesses large mixing couplings of the same order as the diagonal scales ($|H_{rL}| = 177.6$, $H_{L\eta} = 355.3$). Releasing L deepens the existing saddle-point structure: the minimum eigenvalue deteriorates from $\lambda_{\min} = -35.9$ in the (r, η) 2×2 block to $\lambda_{\min} = -349.8$ in the full 4×4 block. Accordingly, L is *not* added to the final mode dictionary (H.3); the treatment and potential stabilisation of L is left as a separate analysis beyond the scope of the present paper.

H.3 Final Mode Dictionary with Numerical Anchors

Variable	Classification	Numerical anchor	Note
q_A	spin-2	$\mu_q^2 = 48\pi^2$ at $\varepsilon = 0$	final basis
X_k	spin-1 physical	$\lambda_{\text{phys}} = 4.1867$	quadratic decoupled
Y_k	spin-1 auxiliary	$GY_k = 0$, $H_{\text{null}} = 894.845$	not gauge
η	torsion scalar	AX control parameter	physical input
V	torsion scalar	VT control parameter	physical input
P	topological density	AX/VT zero; MX nonzero	chirality indicator

Table 13: Final mode dictionary with numerical anchors for paper III.

At the isotropic MX point the torsion scalar block satisfies numerically:

$$H^{(\eta,V)} = \begin{pmatrix} 355.31 & 0 \\ 0 & 355.31 \end{pmatrix}. \tag{106}$$

Before accounting for coupling to the radial mode, the torsion subsector is locally stable and doubly degenerate.

(r, η) **coupled block eigenvalue analysis (AX mode,** $\eta = 1$**,** $V = 0$**):**

$$H^{(r,\eta)} = \begin{pmatrix} H_{rr} & H_{r\eta} \\ H_{\eta r} & H_{\eta\eta} \end{pmatrix} = \begin{pmatrix} \approx 0 & 118.44 \\ 118.44 & 355.31 \end{pmatrix}. \tag{107}$$

Quantity	Value
H_{rr}	≈ 0 ($\approx -10^{-6}$)
$H_{r\eta}$	118.435
$H_{\eta\eta}$	355.306
$\det H^{(r,\eta)}$	-1.40×10^4
λ_1 (minimum)	-35.86
λ_2 (maximum)	391.17

Table 14: Eigenvalue analysis of the (r, η) coupled block in AX mode.

$H_{rr} \approx 0$ indicates that r is in a *flat direction (zero mode)* of the effective potential in the AX background. Consequently $\det H^{(r,\eta)} < 0$ and the 2×2 block is saddle-point type (indefinite). The eigenvector of the negative eigenvalue is predominantly in the r direction ($\approx -0.957\, r + 0.290\, \eta$), so the instability originates in the flat direction of r and its coupling to η. The eigenvalue of the predominantly η direction $\lambda_2 \approx 391$ remains positive.

In the VT mode ($\eta = 0$, $V = 1$), $H_{rr} = 118.44$ (non-zero) and $H_{r\eta} \approx 0$, forming a positive-definite block. Thus the r–η mixing and the r zero mode are specific to the axial torsion component η.

H.4 Representative Phase Atlas on the Axisymmetric Slice

ε	$\mu^2_{m=0}$	$\mu^2_{m=\pm2}$	$\mu^2_{m=\pm1}$	Phase note
-0.40	675.1	-239.8	117.4	$m = \pm2$ branch unstable
-0.30	793.9	-12.5	321.8	near ε_{c-}
-0.20	734.5	203.9	445.9	stable negative side
-0.10	610.9	364.1	487.1	stable negative side
0.00	473.7	473.7	473.7	isotropic Schur point
0.20	219.8	685.1	413.3	positive-side split
0.40	44.9	842.2	359.8	near ε_{c+}
0.482	0.49	891.8	341.6	$m = 0$ soft mode

Table 15: Representative phase atlas on the axisymmetric slice $s = 0$.

The boundary points are:

Boundary	Value	Vanishing branch
ε_{c-}	-0.29309 ± 0.00019	$\mu^2_{m=\pm2} \to 0$
ε_{c+}	0.48324	$\mu^2_{m=0} \to 0$

H.5 Composite Visualization at Selected $H_{\varepsilon\varepsilon} = 0$ Points

The following figures display, side by side, (i) the (ε, η) phase diagram, (ii) the effective potential $V_{\text{eff}}(r)$ at the selected point, and (iii) the three spin-2 spectral branches $\{m = 0,\ m = \pm2,\ m = \pm1\}$. The purpose is to provide a comprehensive view of the relationship between the phase-diagram position, the potential shape, and the spectral splitting analysed in §4 and Appendix D.

A note of caution: $H_{\varepsilon\varepsilon} = 0$ is a *local softening condition along the squash direction* and does not always coincide with the actual phase-transition point.

$\alpha = 0$, Type-I representative point. Baseline softening figure. Corresponds to the positive-side threshold $\varepsilon_{c+} = 0.48324$ of the main text, where the $m = 0$ branch just reaches zero while the $m = \pm2$ branch remains massive.

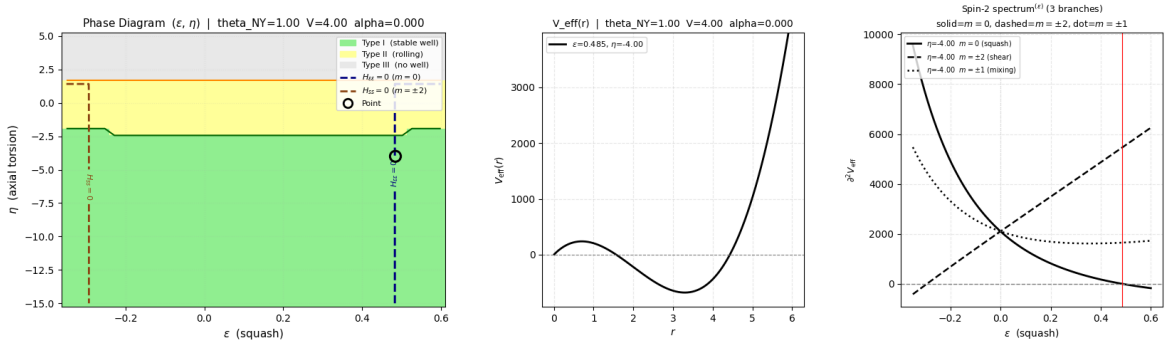


Figure 16: Composite visualization at a positive-ε, $\alpha = 0$ point on the $H_{\varepsilon\varepsilon} = 0$ locus.

$\alpha < 0$, Type-IW/IIW representative point. Example of local softening on the $\alpha < 0$ side. From paper II, $\alpha \leq 0$ is the stable side for the isotropic vacuum, forming the Type-IW and Type-IIW stable phases. The spectral splitting seen here is a normal branch splitting associated with a shift of the background point, not a ghost instability.

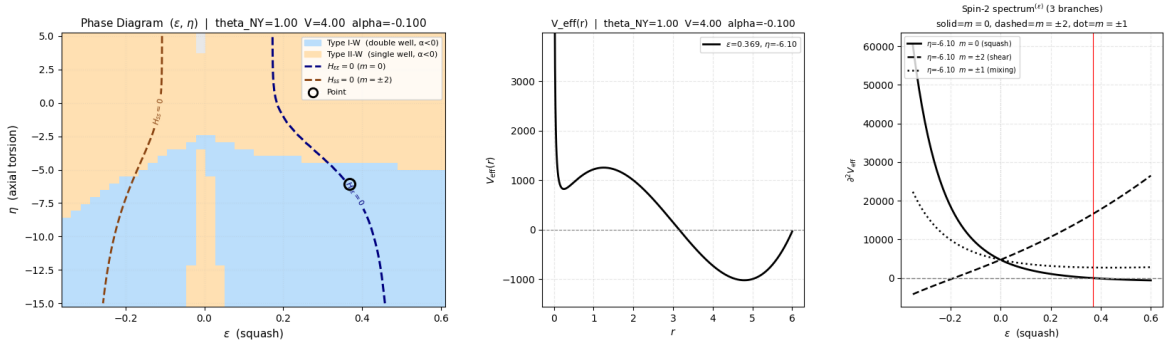


Figure 17: Composite visualization Type-IW at a positive-ε, $\alpha < 0$ representative point.

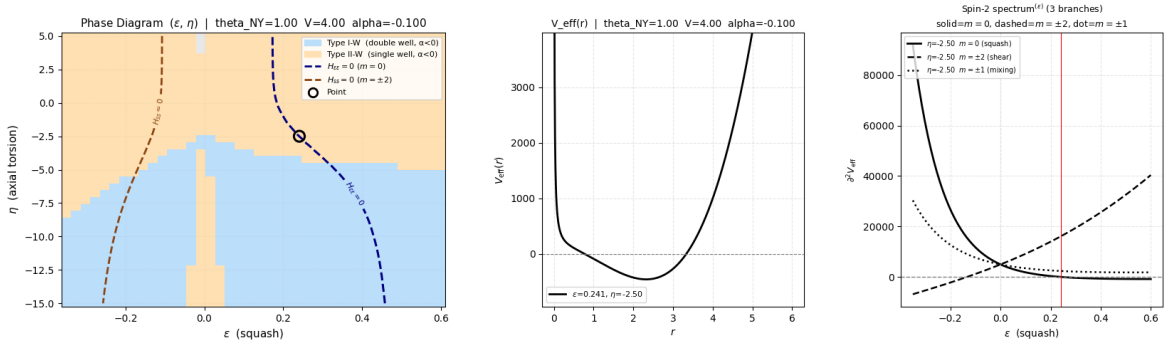


Figure 18: Composite visualization Type-IIW at a positive-ε, $\alpha < 0$ representative point.

The figures function as a supplemental phase atlas that allows simultaneous comparison of the phase classification, the radial potential shape, and which spectral branch participates in the softening.

$\alpha > 0$**, Type-G representative point.** Supplemental visualisation is available on the ghost/metastable side, but the physical interpretation of the spectral branches requires care. In this paper the region where $\alpha > 0$ and the potential may possess a local well is classified as Type-G in the phase diagram. This region corresponds to the ghost-unstable side, which does not yield a globally stable vacuum [6]. Accordingly, $H_{\varepsilon\varepsilon} = 0$ and $H_{ss} = 0$ within the Type-G region should not be interpreted on the same footing as critical conditions in the stable phases; a detailed physical interpretation of these loci is outside the scope of the present paper.

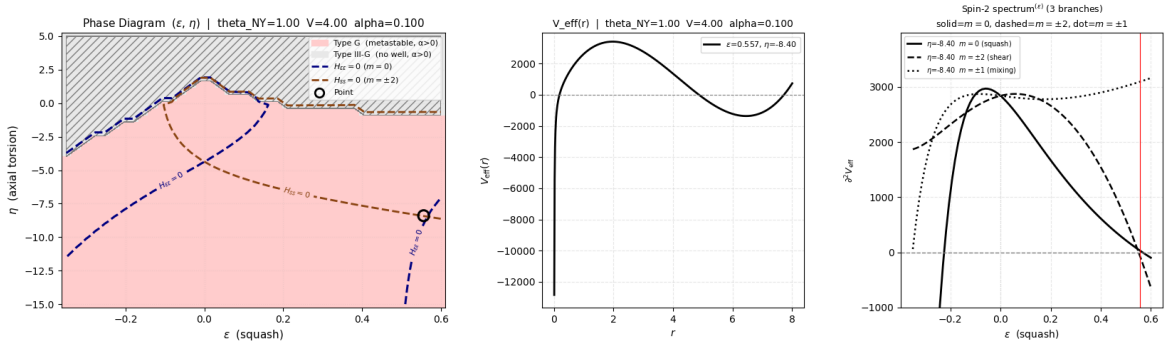


Figure 19: Composite visualization at a positive-ε, $\alpha > 0$ representative point.

H.6 Scripts

The scripts generating the numerical tables in this appendix are:

- `paper03/spin0_torsion_cross_hessian.py`: spin-2 × torsion and radial × torsion cross-Hessians, torsion block $H^{(\eta,V)}$ (§H .2).
- `paper03/spin0_reta_block_eigenvalues.py`: (r, η) coupled-block eigenvalue analysis (§H.3).
- `paper03/spin0_rLetaV_hessian.py`: 4×4 Hessian $H^{(r,L,\eta,V)}$ (§H .2A).

H.7 Remarks

The numerical values in this appendix serve as reference anchors for the stability discussions and mode dictionaries in the main text. The phase atlas here samples representative points only; full-grid supplemental data and high-density figures for publication are deferred to subsequent work or supplemental material as needed.

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